TOPPR PERSONAL STUDY VAULT

Plant Pathology

Section B Overview & Study Guide

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Section Study Notes

Study Notes: Plant Pathology (3201) - Section B (Rezaul Sir)

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Lecture: 1. Rice Diseases

Course Slides: 1. Rice Diseases

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Diseases of Rice

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DISEASES	CAUSAL ORGANISM
FUNGAL DISEASES	
Blast	Piricularia oryzae
Brown spot	Drechslera oryzae/ Bipolaris oryzae
Narrow brown spot	Cercospora oryzae
Stem rot	Sclerotium oryzae
Sheath blight	Rhizoctonia solani
Bakanae or foot rot	Fusarium moniliforme/ Gibberella fujikuroi
Leaf scald	Gerlachia oryzae
False smut	Ustilaginoidea virens
Sheath rot	Sarocladium oryzae
Grain discoloration	Bipolaris oryzae/ Fusarium moniliforme/ Sarocladium oryzae/ Alternaria padwickii/ Phoma sorghina

DISEASES OF RICE

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BACTERIAL DISEASES	
Bacterial leaf blight	Xanthomonas campestris pv. oryzae
Bacterial leaf streak	Xanthomonas campestris pv. oryzicola

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DISEASES CAUSE BY NEMATODE	
Ufra (Dakpora)	Ditylenchus angustus
White tip of rice	Aphelenchoides besseyi
Root knot of rice	Meloidogyne graminicola

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VIRUS DISEASES

(Also a mycoplasmalike organism [MLO] disease)

Tungro (rice tungro disease) (RTD)- Rice tungro spherical virus (RTSV)

Rice tungro bacilliform virus (RTBV)

Rice grassy stunt- Rice grassy stunt virus (RGSV)

Rice hoja blanca- Rice hoja blanca virus (RHBV)

Rice yellow dwarf- MLO

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BLAST OF RICE

Causal organism:

Pyricularia oryzae

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A conidium and conidiogenous cell of M. grisea

[Image not found: /uploads/1. Rice Diseases_s8_img4.jpg]
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Symptoms

[Image not found: /uploads/1. Rice Diseases_s9_img1.png]

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[Image not found: /uploads/1. Rice Diseases_s10_img1.png]

Symptoms

Collar blast

Collar blast occurs when the pathogen infects the collar that can kill the entire leaf blade.

Node blast

The pathogen also infects the node of the stem that turns blackish and breaks easily; this condition is called Node blast.

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[Image not found: /uploads/1. Rice Diseases_s11_img1.jpg]

Symptoms

Leaf:

Spindle or eye shaped lesion, ash colour

in center with dark brown margin.

Under favorable conditions, lesions enlarge

and coalesce eventually killing the leaves.

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[Image not found: /uploads/1. Rice Diseases_s12_img1.jpg]
[Image not found: /uploads/1. Rice Diseases_s12_img2.jpg]

Fig: Leaf Blast

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[Image not found: /uploads/1. Rice Diseases_s13_img1.jpg]

Symptoms

Panicle:

At the base of panicle brown to black spots

or rings are formed called –

“Neck Blast/ Panicle Blast”.

Early neck infection:

The grains do not fill as the food materials can

not reach to the young grains and ear turns

white which is known as white head symptoms.

[Image not found: /uploads/1. Rice Diseases_s13_img2.jpg]

Fig: Neck Blast

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Symptoms

[Image not found: /uploads/1. Rice Diseases_s14_img1.png]

At later stage neck infection:

The grains are filled up partially or incompletely, the rotten neck can not hold them and ear breaks down at the point of infection

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Disease cycle:

1. The blast fungus over winters on infected seed, grasses and

other colateral host (ginger, sedge, etc.)

2. Usually propagated by wind borne spores under humid conditions
3. The conidia falling on leaves and germinates and penetrates into the leaves.
4. The incubation period varies with temperature, within 4-5 days

the symptoms are seen.

5. The secondary infection cased by splash of rain or wind.

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Rotten neck symptom causing break down

Of the panicle at nodal region

Infected seeds,

weeds as primary source of infection

Conidiophores with

Conidia on the lesion

Infected area enlarged

Young infected plant with spindle

Shaped lesions on the leaf

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Factors influencing disease development

High relative humidity (90% or higher)

High soil nitrogen

Long dew period (10hrs or more)

Light rainfall or drizzle, amount of solar radiation & overcast sky

Besides, close planting and low silica content in the soil

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Control measures

Resistant varieties: BR-3, BR-4, BR-6, BR-8, IR-22, etc.

Clean seeds should be sown

Seedlings should be raised under low land condition.

Avoid excess N and add 100 kg of K fertilizer/ha during field preparation

Seed treatment with Granosan–M or Agrosan–GN

@ 2 tolas/maund. Or Vitvax 200 @0.25% of seed weight.

Fungicides such as Benlate/ Topsinmethyl/ Blasticidine/

Hinosan @ 0.2% should be sprayed on the leaves at the time of panicle emergence to prevent neck infection.

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Brown spot of rice

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Very common disease in all rice growing countries.

First described and reported in 1900 as seeding blight.

Appeared in epidemic form and contributed to the Bengal famine of 1943.

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[Image not found: /uploads/1. Rice Diseases_s20_img1.jpg]

Causal organism Bipolaris oryzae/ Drechslera oryzae

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Brown spot

[Image not found: /uploads/1. Rice Diseases_s21_img1.jpg]

Bipolaris oryzae

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Symptoms:

[Image not found: /uploads/1. Rice Diseases_s22_img1.jpg]

Attacks rice plant at all stages of its growth.

On the coleoptile the spots are brown and circular to oval.

One or more spots may coalesce to form bigger discolored zones.

Symptoms of the disease are spots on the leaves and the glumes, may also appear on the leaf sheath, panicles and stems.

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[Image not found: /uploads/1. Rice Diseases_s23_img1.jpg]

Symptoms:

Spots on the leafs first appear as minute deep brown dots.

The spots enlarge and become circular to oval and dark brown.

The spots are surrounded by a yellow halo.

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Symptoms:

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When glumes are infected black spots appears on the surface.

In severe cases the whole surface of the glumes may be coated with dark brown mycelium and spores and form a velvety mat.

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Suitable Factors

Soil with poor nutritional elements and toxic substances

Deficiency of potassium, magnesium and silica

Also appears when nitrogen deficiency are induced at late growth stage

Dry soil, too high or too low water temperature

High relative humidity, continued rain and cloudiness

Seedling infection takes place when soil temperature below 26°C

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Control

Cultivate resistant varieties (Biplob, Brrisail, progoti, mukta, etc.)

Sowing healthy seeds

Burning crop residues

Hot water treatment 50° C for 30 min.

Balance fertilizer

Seedbed and land soil should be kept wetty

Benlate/ Hinosan @ 0.2% at 10-15 days interval from 60 days of seedling

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STEM ROT OF RICE

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Causal Organism:

Sclerotium oryzae

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[Image not found: /uploads/1. Rice Diseases_s32_img1.png]

Lesion of stem rot disease on rice leaf sheath

[Image not found: /uploads/1. Rice Diseases_s32_img2.png]

Rice stem rot symptom and sclerotia of pathogen

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Symptoms

Irregular water soaked areas are appeared on sheath at water line

[Image not found: /uploads/1. Rice Diseases_s33_img1.jpg]

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Symptoms

These areas turn black and become large.

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Symptoms

Excessive production of the late tillers and rotten in severe cases

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Symptoms

Culm and inner sheath are affected.

[Image not found: /uploads/1. Rice Diseases_s36_img1.jpg]

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Symptoms

[Image not found: /uploads/1. Rice Diseases_s37_img1.jpg]

Black powdery sclerotia are found within the culm and leaf sheath.

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DISEASE CYCLE

Black lesion around water line

Minute sclerotia formed inside stem

Sclerotia inside the stem

Sclerotia attack the leaf sheath.

Floating sclerotia in water.

Sclerotia in the soil

[Image not found: /uploads/1. Rice Diseases_s38_img1.jpg]

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SUITABLE FACTORS:

Stagnant water in the field

Excessive nitrogen and low potash fertilizer

Infested with stem borer

[Image not found: /uploads/1. Rice Diseases_s39_img1.jpg]

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CONTROL:

Resistant varieties– Biplob, Borishal, Asha, Progoti, Mukta, etc. should be cultivated

Stubble and residues should be burnt

Crop rotation

Drained out the water and allowing the soil to dry and crack before re-watering

Avoid use of excess nitrogen, balance with K

Spray with Benlate/ Homai/ Topsin-M at the booting and before flowering stage @ 0.2 % at 9-15 days interval

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Sheath blight of rice

Causal organism:

Rhizoctonia solani

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Sheath blight disease of rice plant

[Image not found: /uploads/1. Rice Diseases_s42_img2.png]

Sclerotia of rice sheath blight disease

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[Image not found: /uploads/1. Rice Diseases_s43_img1.jpg]

Symptoms:

Spots or lesion near water level

Lesions are oval or elliptical

Mature lesions are grayish-white

Margin is dark coloured

[Image not found: /uploads/1. Rice Diseases_s43_img2.jpg]

Infected sheath

Bigger lesions

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Symptoms:

[Image not found: /uploads/1. Rice Diseases_s44_img1.jpg]

Brownish sclerotia form on lesion

Look like skin of cobra snake

[Image not found: /uploads/1. Rice Diseases_s44_img2.png]

Older lesions on sheath

Sclerotia of rice sheath blight disease

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Disease cycle

Over wintering sclerotia in the soil

Sclerotia come in contact with the rice plant

Sclerotium germinating & infecting host

Diseases developing in horizontal & vertical direction

Characteristic lesion develop on the sheath

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[Image not found: /uploads/1. Rice Diseases_s47_img1.jpg]

Fig: Diseases cycle of sheath blight of Rice

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Suitable factors:

High humidity & warm temperature

Close planting

High nitrogenous fertilizer more susceptible to diseases

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Control measures:

The disease is hard to control

Varities with moderate level of resistance include BR-4, BR-5, IR-26, BAU-63, etc.

Spray with Hinosan/ Topsin/ Benlate @ 0.2% at 10-15 days interval

Balance fertilizer and spacing

Destroy crop residues

Dry the field and re-water

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BAKANAE/ FOOT ROT DISEASE OF RICE

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CAUSAL ORGANISM

Fusarium moniliforme

Gibberella fujikuroi (perfect stage)

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[Image not found: /uploads/1. Rice Diseases_s51_img2.jpg]

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[Image not found: /uploads/1. Rice Diseases_s52_img1.png]

Microconidia and macroconidia of rice bakanae disease pathogen

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SYMPTOMS

[Image not found: /uploads/1. Rice Diseases_s53_img1.png]

Infected plants are several inches taller than normal plants in seedbed and field

Plants are slender, thin and yellowish green in color

[Image not found: /uploads/1. Rice Diseases_s53_img2.jpg]

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[Image not found: /uploads/1. Rice Diseases_s54_img1.jpg]
[Image not found: /uploads/1. Rice Diseases_s54_img2.jpg]

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SYMPTOMS

Plants produce adventitious roots at the lower node of the culms.

Diseased plants bear few tillers and leaves dry up quickly. The affected tillers usually die before reaching maturity.

Partially filled grains, sterile, or empty grains for surviving plant at maturity

[Image not found: /uploads/1. Rice Diseases_s55_img1.jpg]

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SUITABLE FACTORS

Soil temperature of 35° C seedling infection (no-of 20° C or below)

Damp soil condition

High nitrogenous fertilizer application

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DISEASE CYCLE

[Image not found: /uploads/1. Rice Diseases_s57_img1.jpg]

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CONTROL MEASURES

Collection of seed from disease free area

Destroy crop residues

Seed treatment with Vitavax-200/ Granosan-N/ Brassicol/ Homai (1: 1000)

Balance fertilizer should be used

Uprooting and destroying of the infected plants

Resistant variety- Biplob, Brrisail, Progati, Mukta, etc.

Foot rot- dry

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Bacterial leaf blight

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Bacterial leaf blight

Xanthomonas campestris pv. oryzae

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[Image not found: /uploads/1. Rice Diseases_s61_img2.jpg]
[Image not found: /uploads/1. Rice Diseases_s61_img3.jpg]

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SUITABLE FACTORS

High temperature

High nitrogenous fertilizer

Potassium and phosphorous deficiency

Excess silicon and magnesium

Heavy rain flooding

Stagnancy of water and severe wind

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Disease cycle

It may survive on rice straw/ seed

The initial infection may occur in nursery but the seedlings may not show the disease.

Insect or from chaffing or striking of leaf against each other in the wind and initiated infection.

Bacteria after air borne/ irrigation water are carried from one plot to another.

The healthy leaves may get the contamination by contact or by rubbing with disease leaf.

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CONTROL MEASURES

Plant residues should be burnt

Seed should be collected from healthy plants

Agrimycine should be sprayed as soon as disease appear

Resistance varities should be cultivated such as

IR-20, IR-22, BR-4, BR-14, BAU-63, etc.

Balanced fertilizer should be used

Field should be dried and allow to crack

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Ufra Disease of rice

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Causal organism:

Ditylenchus angustus

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Symptoms:

[Image not found: /uploads/1. Rice Diseases_s67_img1.jpg]

The first symptoms are cholorosis with white patches or striping of the uppermost leaves of rice plants which are more than two months old.

The leaves including flag leaf and panicle can be twisted and deformed with a frequently wavy margin.

[Image not found: /uploads/1. Rice Diseases_s67_img2.jpg]

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Symptoms:

[Image not found: /uploads/1. Rice Diseases_s68_img1.jpg]

In severe attack ,two distinct types of symptoms are noted on the panicle –

when the panicle remain s largely enclosed inside the boot, while the stalk partly emerges and shows branching, it is called Swollen or Kutcha or Thor ufra.

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Symptoms:

when the panicle emerges it gets distorted with a dark brown discolored peduncle and produces normal grains only near the tip but the grains below are sterile and shrivelled. it is called Ripe or pucca ufra.

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[Image not found: /uploads/1. Rice Diseases_s70_img1.jpg]
[Image not found: /uploads/1. Rice Diseases_s70_img2.jpg]

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Disease cycle

[Image not found: /uploads/1. Rice Diseases_s71_img1.jpg]

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Disease cycle

1. Obligate ecto-parasite
2. Remain dormant in stubbles and crop residues
• coiled in a circular fashion
3. Placed in water they become active and uncoil
4. Climb up and enter the growing tissue
5. Suck the sap from the epidermal cells with their stylets
6. Multiply and move up to new tissue

-plant to plant and plot to plot

7. Harvest again go for dormancy

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[Image not found: /uploads/1. Rice Diseases_s73_img1.jpg]

Fig : Ditylenchus angustus (stem or ufra nematode)

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Suitable factors:

Infections occur best at 28-30’ C and near saturated RH (>75’c)

The diseases is most severe in the wettest areas and in the fields with highest water level.

Ufra infected rice is more susceptible to the blast pathogen, and injuries caused encourages a few more fungi.

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Control measures

Cultural control:

Burning or removal of crop residues

Summer ploughing

Destruction of infested seedling in the seedbed

Crop rotation (jute+deep water rice, mustard+low land rice etc very effective)

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Control measures

Chemical control

1. Diazinon at 100 ppm can be sprayed to completely eradicate the nematode in 72 hours

2. Seed treatment with carbofuran at 0.3 kg a.i./80 kg

Carbosulfan, benomyl, ethoprophos also effective

3. Spreading of Furadan granules 5kg/bigha to reduce nematode population

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Control measures

Resistant varieties

Cultivate resistant variety such as Raida, Bajail-65 or early variety like Digha

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DISEASE NAME: TUNGRO OF RICE CAUSAL ORGANISM: RICE TUNGRO VIRUS

[Image not found: /uploads/1. Rice Diseases_s78_img1.jpg]

Fig: Tungro in the field

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DISEASE NAME: TUNGRO OF RICE CAUSAL ORGANISM:

RICE TUNGRO VIRUS

Transmitted by

Green leaf hopper: Nephotettix virescence

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INTRODUCTION

Tungro disease has been known to occur in Philippines for many years . It also occurs in Indonesia and many others countries. In Bangladesh this disease is recorded in 1966 and now it has become one of the major disease of rice.

[Image not found: /uploads/1. Rice Diseases_s80_img1.png]

Fig: Tungro in the field

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SYMPTOMS

[Image not found: /uploads/1. Rice Diseases_s81_img1.png]

The plants become stunted.

Number of tillers are reduced and the leaves become yellow.

Tillering is influenced by the age of the infected plants.

Fig: tungro of rice

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SYMPTOMS

Yellowing ranges from light yellow to orange yellow or brownish yellow starts from the tip of the leaves.

Pale yellow stripes are found.

The stripes are parallel to the veins.

[Image not found: /uploads/1. Rice Diseases_s82_img1.jpg]

Fig: Leaf symptoms of Tungro (IRRI)

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SYMPTOMS

The yellow orange color persists for sometimes and the leaves die.

Infected plants bear a few poor panicles.

[Image not found: /uploads/1. Rice Diseases_s83_img1.jpg]

Fig: Leaf symptoms of Tungro (IRRI)

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SUITABLE FACTORS

[Image not found: /uploads/1. Rice Diseases_s84_img1.jpg]

Presence of the virus sources

Presence of the vector

Age and susceptibility of host plants

All growth stages of the rice plant specifically the vegetative stage

Fig: Green leaf hopper

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LIFE CYCLE:

Tungro is transmitted by the rice green leaf hopper (Nephotettix virescence).

The insect acquires the virus by feeding on a diseased plant for 30 minutes.

The virus has no incubation period in the insect.

The insect become infective for a short time.

[Image not found: /uploads/1. Rice Diseases_s85_img1.jpg]

Fig: Green leaf hopper

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LIFE CYCLE:

The tungro virus does not persist in the vectors.

Infective nymph does not retain its infectivity after moulting .

It may become infective when it again feeds on diseased plants.

Symptoms generally appear about a week after inoculation feeding.

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[Image not found: /uploads/1. Rice Diseases_s88_img1.jpg]

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CONTROL

Resistant varieties such as Mala, Brisail, Nazersail, Kalojira, BR-5,8,10,12 etc. should be cultivated.

Weeds and diseased plants Should be eliminated.

Since the virus is transmitted by leaf hopper the disease can be reduced by insect control with insecticides.

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CONCLUSION

Tungro is one of the most damaging and destructive diseases of rice in countries in Southeast Asia. Outbreaks of the disease can affect thousands of hectares in many countries. Plant infected with the virus at the early crop growth stage could have as high as 100% yield loss in severe cases. So we should control the rice tungro disease.

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Diseases of Rice

Bacterial Leaf Streak of Rice.

Narrow Brown Leaf Spot of Rice.

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Bacterial Leaf Streak of Rice

Causal Organism

Xanthomonas campestris pv. oryzicola

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Symptom- 01

Initially water-soaked fine, interveinal, long/ short streaks/lesions found on the leaves. water-soaked streak

[Image not found: /uploads/1. Rice Diseases_s93_img1.jpg]

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Symptom- 02

Lesions become enlarge gradually, borders turn brownish and the centre remain gray.

The lesions give centre translucent against the sun. border

[Image not found: /uploads/1. Rice Diseases_s94_img1.jpg]

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Symptom- 03

On lesions- minute yellowish beads of dried bacterial exudates may be found. yellowish streak

[Image not found: /uploads/1. Rice Diseases_s95_img1.jpg]

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Symptom - 04

A large portion of leaf blade becomes yellow which is confused with tungro. yellowing

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Disease Cycle

Overwinter- in the seed, straw and weeds.

Primary infection- occur in the nursery bed but seedlings not show the disease until a few weeks after transplantation.

Secondary infection- the bacteria enter the leaf through stomata and wounds.

The disease transmitted by winds and also by irrigation water.

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Fig. Of Disease Cycle

[Image not found: /uploads/1. Rice Diseases_s98_img1.jpg]

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Factors Favoring Disease Development

High temperature (about- 300)

High relative humidity

Dense tillering

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Control Measures

Burn rice straw and debris.

Collect seeds from disease free area.

Use resistant varieties BR-4, BR-11, BR-14, IR-20, IR-22, BAU-63 etc.

Use balanced fertilizer (20Kg/ bigha).

Spray agrimycin.

Treat seeds with agrimycin @ 0.25%.

Dry the field and allow to crack before irrigation.

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Narrow Brown Leaf Spot of Rice

Causal Organism

Cercospora janseana

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Symptom- 01

symptoms usually observed during the later growth stages.

short, linear, brown lesions found on the leaves and also on leaf sheaths, pedicels, and glumes. lesion

[Image not found: /uploads/1. Rice Diseases_s102_img1.jpg]

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Symptom- 02

The lesions are 2 to 10 mm long and 1 mm wide. lesion

[Image not found: /uploads/1. Rice Diseases_s103_img1.jpg]

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Symptom- 03

A net blotch - like pattern forms on leaf sheaths. net blotch

[Image not found: /uploads/1. Rice Diseases_s104_img1.jpg]

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Disease Cycle

[Image not found: /uploads/1. Rice Diseases_s105_img1.jpg]

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Conidia of Narrow Brown Leaf Spot Disease Pathogen

The conidiophores are emerging from stomata, solitary or in group of two or three, dark, paler at apex, three or more septate.

[Image not found: /uploads/1. Rice Diseases_s106_img1.png]

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Factors Favoring Disease Development

temperature from 25-28° C

Potassium deficiency.

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Control Measures

Cultural practices- a) The use of balanced (K & P) fertilizers. b) The use of early maturing varieties. c) The use of resistant varieties .

Spraying of fungicides- benomyl, iprodione, propicanazole etc.

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Root Knot of Rice

Root knot is a very common disease of rice and occurs in most rice growing countries. More than 30- 70% yield loss in rice occurs due to root knot disease. It is regarded mainly as a problem of nursery bed & it has frequently been noticed in the main fields as well.

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Root Knot of Rice

The disease is caused by:

Meloidogyne graminicola

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Root Knot of Rice

[Image not found: /uploads/1. Rice Diseases_s111_img1.jpg]

Factors influencing disease development :

18-23.5 0c for root knot development

18.5-20.50c for egg mass production

Stagnant water

Sandy soils with particles>30 micro meter in diameter.

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Root Knot of Rice

Symptoms :

Above ground symptoms

Reduced growth, small pale green or yellowish leaves tend to wilt.

Affected plants – stunted, become yellow in color.

Reduced tillering

Leaves – curl along mid rib, dried & turn to bronze color.

Flowering delayed & poor.

[Image not found: /uploads/1. Rice Diseases_s112_img1.jpg]

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Root Knot of Rice

Symptoms :

Below ground symptoms

Root tip growth retarded.

Most diagnostic symptom- presence of root gall due to hypertrophy & hyperplasia.

[Image not found: /uploads/1. Rice Diseases_s113_img1.jpg]

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Root Knot of Rice

[Image not found: /uploads/1. Rice Diseases_s114_img1.jpg]

Disease cycle

Over wintering : plant debris, soil, alternate hosts (Portulaca oleracea, Amaranthus viridis etc).

Infective 2nd stage juveniles enter root tips – primary inoculum.

Within 2-3 days, juvenile become established.

Cells around heads enlarged due to saliva secretion by nematode during feeding.

The nematode undergo moulting- become adult.

Slide 115

Root Knot of Rice

Disease cycle

The adult female produces egg sacs & gives rise to vast number of juveniles- source of secondary inoculums.

The juvenile attack adjacent root tip – secondary inoculum.

Nematode become coiled, fall on soil, washed out with irrigation water, disseminated from plant to plant, plot to plot & the cycle proceeds repeatedly.

[Image not found: /uploads/1. Rice Diseases_s115_img1.jpg]

Slide 116

Root Knot of Rice

Control

Collect seeds from disease free sources.

Resistant variety- TKM- 6, Patnai 6, Garem, Hamsa etc.

Tolerant variety- Basant Bahar.

Destroy crop residue

Flooding, fallowing, rewatering.

Destruction of alternate host.

Crop rotation.

Seed treatment- Thionazin, Cereson dry @ 0.02% of seed weight .

Sterilization of soil or soil fumigation.

[Image not found: /uploads/1. Rice Diseases_s116_img1.jpg]

<div id="2-field-crop-diseases-wheat-maize-barley-jute-cotton-sugarcane-groundnut-"></div>

Lecture: 2. Field Crop Diseases (Wheat, Maize, Barley, Jute, Cotton, Sugarcane, Groundnut)

Course Slides: 2. Wheat, Maize, Barley, Jute, Cotton, Sugarcane, Groundnut

Slide 1

Loose Smut of Wheat

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s1_img1.jpg]

Slide 2

Loose Smut of Wheat

Causal organism: Ustilago tritici

Slide 3

Symptoms:

Symptoms appear after ear emergence

Diseased ear emerged first

Black powdery mass are developed

Except awns all parts of ear converted into smut spores

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s3_img1.jpg]

Slide 4

Symptoms cont……..

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s4_img1.jpg]

Black power in ear- covered by silvery membrane

Membrane burst later and smut spore release

Smut spores blowing by wind and central axis (rachis) remain

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s4_img2.jpg]

Slide 5

Suitable factors:

Favorable Temperature: 23 ■ C

For the dispersal of smut spore: 16-22 ■ C

Relative Humidity: 60-85%

Wind and light showers or heavy dew

Slide 6

Disease cycle:

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s6_img1.jpg]

Slide 7

Control Measures:

Use of disease free seeds

Release of disease resistance varieties (BARI Gom-25, BARI Gom-26, BARI Gom-27, BARI Gom-28, BARI Gom-29 BARI Gom-30, BARI Gom-31)

Seed treatment with Vitavax-200 at the rate of 0.25% of seed weight

Spray Bavistin-50 WP at the rate of 0.1% at 2-3 times in every 7-10 days interval

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Slide 8

Control Measures cont…..

Slide 9

Leaf blight of Maize

Slide 10

CAUSAL ORGANISM:

Drechslera turcicum

Slide 11

SYMPTOMS:

All parts but conspicuous lesions on leaves.

Begins small yellowish, round or elliptical spots on leaves.

Extends along the leaf and may reach 5 inches or more in length and 1 or more inches in width.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s11_img1.jpg]

Slide 12

Edges of lesions are usually well-defined & infected areas become tan to brown.

Tissues become thin & semi transparent.

Centre of the lesions brownish black dusty (for spore).

Slide 13

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s13_img1.jpg]

Infected leaves gradually dry up & assume burnt appearance.

Severe infection- death of leaves and yield loss resulting from incomplete filled grain.

Slide 14

SUITABLE FACTORS :

Conditions that favours the growth and development of the plant rapidly.

Moist condition for normal growth of plant is also sufficient for spore germination and infection.

Slide 15

DISEASE CYCLE

The fungus attacks many grasses & sorghum.

The primary infection originates from spores carried by the wind from weed hosts or crop residues.

Slide 16

The spores germinate in the whorls of the plant and may produce several lesions.

The spores produced on these lesions helped in secondary spread of the disease.

Slide 17

CONTROL

Dead leaves, stubbles & weed hosts should be destroyed after harvesting.

Seed should be collected from healthy plants.

Slide 18

Resistant verities should be cultivated.

Crop rotation may be done for reducing disease intensity.

Spray with Tilt 250 EC @ 0.04% at an interval of 15 days (Captan or Znieb also effective).

Slide 19

Covered Smut Of Barley

Slide 20

Causal Organism:

Ustilago hordei

Slide 21

Symptoms

Become apparent when the ears emerge, with every spikelet transformed into a sorus.

At first the sori are covered with a thick membrane, which resists easy rupturing.

The kernels are replaced by black spore mass.

Infected heads can emerge later than healthy heads.

Infected plants are shorter.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s21_img1.png]

Fig; Covered Smut Of Barley

Slide 22

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s22_img1.jpg]

Fig; Covered Smut of Barley

Slide 23

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s23_img1.png]

Disease cycle of covert smut or bunt of barley

Slide 24

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s24_img1.jpg]

Slide 25

Disease cycle

Seedling infection initiates the disease. Teliospores penetrating in the soil or seed surface from spordia which infect the seedlings.

Establish dikaryotic mycelium that grows systematically in the host.

In the ovary, the mycelium is transformed into teliospores, which adhere to seed surface or fall on the ground and penetrate when the next crop is shown.

Slide 26

Suitable factors

Infection occurs as the seedlings emerges from the sprouting seed and is favored by cool soil temperatures (10-13° C).

Slide 27

Control

Seed treatment with Vitavax-200 @ 0.25% of seed weight or Agrosan GN @ 2-2.5gm/kg seed.

Disease free seed should be collected.

Resistant varieties.

Slide 28

Stem rot of Jute

Slide 29

Causal Organism: Macrophomina phaseolina

Slide 30

Symptoms:

Plants are attacked at all stages of growth

Seedling:

blackish streaks appear on hypocotyls & cotyledons, in moist condition■ damp■off.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s30_img1.jpg]

Slide 31

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s31_img1.jpg]

Mature plants:

leaves are infected & develop to black lesion ■along apex & margin, mid rib and petioles

dark brown■black lesion at the nodal region ■spread along stem

cortex■ becomes shredded and expose fibrous tissue

Slide 32

plant shade leaves, stem rot & the plant die ultimately

severe cases■ capsule & seeds are affected

numerous pycnidia develop on diseased parts

Slide 33

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s33_img1.jpg]

Stem rot infected jute plot

Slide 34

Suitable factors

Moist weather & warm temperature (30■c)

Excess N & low K fertilizer

Slide 35

Disease cycle

Crop residues, soil, seeds

Germinate and attack plant tissues and cause primary infection

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Slide 36

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s36_img1.jpg]

Slide 37

Disease cycle

Pycnidiospores are produce and cause secondary infection.

They carried by wind to leaves and then spread from leaves to steam.

Later on fungus produces sclerotia and over winters in the soil and attack plant in the next season.

Slide 38

Control

Collect seeds from disease free area

Crop rotation

Application of balanced fertilizers (N, K)

Destroy crop residues

Slide 39

Treat the seeds with Vitavax 200/ Homai @ 0.25% of seed weight

Spray Dithane M-45 @0.2% at intervals of 15 days

Resistant variety:

C. capsularies: D■154, CVL■ 1, CVE■3, CC■45

C. olitorius: O■4, O■9897

Slide 40

Black Band of Jute

Slide 41

Name of the Disease Black Band of Jute

Causal Organism

Botryodiplodia theobromae

Slide 42

Symptoms

. Dense black band found around the stem at about 2-3 feet above the ground.

Regular margin

Slide 43

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s43_img1.jpg]

Black Band of Jute

Slide 44

Symptoms

Fiber turns brown and dry

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s44_img1.jpg]

Slide 45

Symptoms

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s45_img1.jpg]

Dense black band found around the stem at about 2-3 feet above the ground.

Regular margin.

The bark of the main stem splits longitudinaly at the later stage.

The fiber turns brown and dry.

Loses all leaves.

The pathogen produces minute black pycnidia on the band.

After rubbing the finger will be black colour.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s45_img2.jpg]

Slide 46

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s46_img1.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s46_img2.jpg]

Black Band

Stem rot

Slide 47

Suitable Factor

Same as Stem rot of Jute

Moist weather and warm temperature (>300 C)

Excess nitrogen and low potash.

Slide 48

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s48_img1.png]

pycnidiospores- double celled

Slide 49

Disease cycle

Slide 50

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s50_img1.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s50_img2.png]

Conidiophore

Mature conidia

Pycnidia

Slide 51

Control measure

Collection of seeds from healthy plants

Seed treatment by Vitavex / Homai @0.25% of the seed weight.

Crop rotation reduces the disease severity.

Balanced fertilizer should be used.

Resistant variety.

Corchorus capsularis: D-154,CVL-1,CVE-3,CC-45

Corchorus olitorius: O-4, O-9897

Fungicide may be spread at an interval of 15 days as soon as the disease appears in the field.

Slide 52

Anthracnose

of Jute

Slide 53

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s53_img1.png]

CAUSAL ORGANISM

Colletotricum corchorum

Slide 54

Slide 55

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s55_img1.jpg]

Anthracnose of jute

Slide 56

SYMPTOMS

The disease manifest itself first as yellowish-brown, water soaked & depressed spots on the stem.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s56_img1.jpg]

Slide 57

The color of the spots may become dark brown & finally black.

Several spots may coalesce to form large patches, girdling the stem.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s57_img1.jpg]

Slide 58

In most cases the fungus invades the vascular bundles.

The lesion develops cracks in the centre & the fibres are exposed, called cankerous symptoms.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s58_img1.jpg]

Slide 59

The pathogen also infects the pods producing necrotic lesions.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s59_img1.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s59_img2.png]

Fig: Infected stem & pod

Fig: Accervuli & conidia

The accervuli are produced on the spots in high RH.

Slide 60

DISEASE CYCLE

Anthracnose is a seed borne disease.

Fungus survives on seed & crop debris.

In suitable environment they infect succeptable host.

They affect stem & pods by producing accervuli.

Thus spores remain in seeds & crop debris, overwintering in it & continue their disease cycle.

Slide 61

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s61_img1.png]

Slide 62

SUITABLE FACTORS

High humidity (over 90%).

Hot & moist weather (July-August).

Slide 63

CONTROL MEASURES

Cultivation of resistant varieties.

Corchorus capsularis: D-154, CVL-1, CVE-3, CC-45

Corchorus olitorius: O-4, O-9897

Slide 64

Rotation with Corchorus olitorius.

Collection of seed from disease-free area.

Destroy & burn crop residues.

Seed treatment with a seed-dressing fungicide like Benlate @ 0.2%.

Slide 65

Jute mosaic virus

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s65_img1.jpg]

Slide 66

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s66_img1.jpg]

Mosaic infected CVL-1 leaves

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[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s67_img1.jpg]

Mosaic infected OM-1 leaves

Slide 68

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s68_img1.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s68_img2.jpg]

Severely infected roots of jute with Nematode

Slide 69

Angular leaf spot of cotton

Occurs through out the world

Reduced yield from shedding of the infected leaves.

Slide 70

Causal Organism: Xanthomonas malvacearum

Slide 71

Symptoms:

Lesions first appear as small water soaked areas.

Enlarge and turn brown to black with age.

On cotyledons round to irregular lesions appear.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s71_img1.jpg]

Slide 72

Lesions developed along leaf veins and leaf petioles.

Lesions are limited by veinletes.

Infected leaves turn yellow and fall to the ground

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s72_img1.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s72_img2.jpg]

Slide 73

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s73_img1.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s73_img2.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s73_img3.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s73_img4.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s73_img5.jpg]

Slide 74

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s74_img1.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s74_img2.jpg]

Slide 75

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s75_img1.png]

Black elongated lesions develop on the young stem. That causes girdling and death of the stem (Black arm).

Slide 76

Boll:

• lesions are round to irregular and become sunken.

Invade the lint and completely destroy the boll.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s76_img1.jpg]

Slide 77

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s77_img1.jpg]

Slide 78

Disease Cycle

Drainage, blowing rain, sprinkler and surface irrigation and field equipments carry the bacteria.

6

Plasmolyze the cell and cause necrosis.

5

Fuzz seed contain bacteria and plant debris.

1

Grow intercellularly and dissolved away the middle lamella.

4

Bacteria enter the plant tissue

2

Bacteria Multiply in sub stomatal chamber.

3

Fig: Disease Cycle of Angular leaf spot of cotton.

Slide 79

Suitable factors:

- Factors that favor the crop, also favor this disease.

Slide 80

Control

Seed from disease free field.

Decomposition of infected plant debris.

Crop rotation

Cultivate resistant varieties

Defoliation with chemicals to reduce late boll infection

Slide 81

RED ROT Of Sugarcane

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s81_img1.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s81_img2.png]

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[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s82_img1.png]

Common Name

Sugarcane red rot

Scientific Name

Colletotrichum fulcatum;

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[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s83_img1.png]

Introduction

Red rot is characterized by interrupted red and white patches within the cane and a sour alcoholic odor when the cane is split open.

The pathogen attacks sucrose accumulating parenchyma cells.

Slide 84

Suitable Factor

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s84_img1.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s84_img2.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s84_img3.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s84_img4.png]

High temperature (30 degree C)

Low pH (5-6)

High RH(90%)

Ratooning of crop

Time of attack

The red rot attack starts from September to October.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s84_img5.png]

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s84_img6.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s84_img7.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s84_img8.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s84_img9.png]

Slide 85

Symptom

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s85_img1.png]

Stalk Symptom

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s85_img2.png]

Leaf Symptom

Slide 86

Stalk Symptoms

Loss of color and drooping of third or fourth leaves from the top are.

The entire top withers.

Canes become shriveled, longitudinally wrinkled.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s86_img1.png]

Slide 87

Stalk Symptoms

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s87_img1.png]

If split open longitudinally, the pith looks red colored.

In advanced stage, red colour may be replaced by dirty brown.

The juice gives a bad smell.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s87_img2.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s87_img3.png]

Slide 88

Leaf Symptoms

Thin reddish lesion on the upper surface of leaves.

Minute red dot on upper surface of midrib of leaf.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s88_img1.png]

Slide 89

Spread of Disease

Primary transmission through soil and diseased setts, while the secondary transmission through air, rain splash and soil.

It is a soil and sett borne disease

Infected propagules in the soil may br conidia, setae, thick walled hyphae.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s89_img1.png]

Slide 90

Disease Cycle

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s90_img1.png]

Slide 91

Disease Management

Disease can be controled by two method.

Cultural Mathod

Select healthy plants for setts

Crop rotation

Resistant and moderately resistant varieties , as CO 244, CO 62399, CO 62101, CO 1336 etc.

Removal of the affected bunches at an early stage

Diseased leaves and canes should be destroyed by burning.

Ratooning should be discouraged.

Slide 92

Chemical Method

Sett should be treated with hot water at 52 degree C.

Setts are dipped in ethylmethylchloride (0.25%) for 5-10 min.

Slide 93

MOSAIC OF SUGARCANE

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s93_img1.jpg]

Slide 94

Causal Organism:

Sugarcane Mosaic Virus Sugarcane mosaic virus is-

A plant pathogenic virus.

Family Potyviridae.

650-770nm long and 12-15nm width with ssRNA genome.

First noticed in Puerto Rico in 1916.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s94_img1.png]

Fig: Microscopic View Of SCMV

Slide 95

Symptoms of Sugarcane Mosaic Virus:

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s95_img1.png]

On the basal region of younger leaves, it is more visible than on older foliage.

Yellowish strips alternate with the normal green portions of the leaf giving the mosaic pattern.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s95_img2.jpg]

Slide 96

Symptoms…

When immature, infected leaves are held up against bright light, yellowish spots with irregular stripes can be seen.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s96_img1.jpg]

Fig: Symptom Level On Leaf

Slide 97

Symptoms…

In severe infections, the yellow strips or necrotic lesions appear on the leaf sheath and stalk and and stem splitting occurs.

Sometimes tip of leaves are abnormally twisted.

The necrotic lesions also develop on the internodes and the entire plant becomes stunted and chlorotic.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s97_img1.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s97_img2.png]

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How The Disease Spreads:

Aphids (Melanaphis sacchari, Rhopalosiphum maidisin etc.) in a non-persistent manner.

Plant infected cane.

Gramineae hosts.

Transferring infected leaf extracts onto healthy leaves.

Transmitted over a short distance by machines and cutting tools.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s98_img1.png]

Fig: Aphids

Slide 99

Suitable Factors:

High vector population.

Using infected setts.

Cultivation of susceptible varieties.

Allow multiple ratooning.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s99_img1.png]

Fig: Vector Population On Sugarcane Leaf

Slide 100

Disease Cycle:

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s100_img1.png]

Slide 101

Disease Cycle…

The virus is mainly transmitted through infected canes used as sett.

The virus also infects Zea mays and a number of other cereals which serve as potential sources of virus inoculum.

The virus also spreads through aphids in a non-persistent manner.

The virus is also sap-transmissible.

The incubation period varies from 7 to 20 days.

Slide 102

Control Measures:

Use disease free setts.

Aerated Stem Therapy(AST) at 56°C for 3 hours, for setts before planting.

Apply malathion @ 0.2% to control the insect vector population.

Uproot and destroy the infected plants.

Avoid multiple ratooning of the affected crop.

Slide 103

Control Measures…

Grow mosaic-resistant or, at least, tolerant varieties.

Breeding mosaic-resistant varieties is needed.

Saccharum spontaneum L. and S. barberi carry resistance to mosaic and so varieties with this background must be preferred.

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s103_img1.png]

Fig: Field Performance of Transgenic Sugarcane Lines Resistant to SCMV

Slide 104

Leaf Spot Of Groundnut (Tikka)

Slide 105

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s105_img1.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s105_img2.jpg]

Introduction

Groundnut (Arachis hypogaea) is one of the important oilseeds and food crops.

Early and late leaf spots, commonly called as tikka disease - economically important foliar fungal diseases.

Infection causes early death of the leaves and dramatic yield loss estimated to range from 10% to 80%.

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CAUSAL ORGANISMS

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s106_img1.jpg]

Early Leaf spot

Late Leaf spot

Slide 107

COMPARISON OF PATHOGENS CHARACTERS

CHARACTERS	Cercospora arachidicola	Cercospora personata
HAUSTORIA	lack	branched
CONIDIOPHORE	1 -2 septate	aseptate
CONIDIA	1-12septate	3-4 septate

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s107_img1.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s107_img2.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s107_img3.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s107_img4.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s107_img5.jpg]

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Symptoms of Early Leaf Spot

Mycosphaerella arachidis (Cercospora arachidicola)

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s108_img1.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s108_img2.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s108_img3.jpg]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s108_img4.jpg]

01

02

03

04

Sub circular to irregular shape

Light brown on lower leaf surface

Redish brown to black on upper leaf(Surrounded by yellow halo)

Sporulation mainly occurs on upper leaf surface and turn to reddish brown colour. Occurs 3-4 weeks after sowing.

Slide 109

Symptoms of Late Leaf Spot

Cercospora personata

Add title

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s109_img1.png]

Smaller spots and more nearly circular than early leaf spot.

01

02

Dark grey or black on lower leaf surface.

No yellow halo around them.

03

Concentric rings of conidia visible on the lower surface.

04

Occurs 5-7 weeks after sowing.

05

Slide 110

Difference Between the Symptoms

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s110_img1.png]
[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s110_img2.png]

Early Leaf Spot

Late Leaf Spot

Infection starts about a month

after sowing

Infection starts around 5 – 7 week

after sowing.

Not perfectly circular reddish brown large spot surrounded by yellow halo.

Mostly circular dark brown small spots without yellow halo.

No larger spot

Larger spot.

On lower surface lesion light brown colour.

Lower surface lesion carbon black colour

Slide 111

Symptoms

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s111_img1.jpg]

Slide 112

Disease Cycle of Tikka Disease of Groundnut

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s112_img1.jpg]

Slide 113

Add title

Favourable Conditions

High Moisture

Above 95%

High Temperature

(Above 25°C)

Close Planting

Dew on Leaf

Surface.

Prolonged Low

Temperature

Slide 114

Control Measure

Previous crop residues should be burnt.

Spray Diathen M-45 (.2%), Bordeaux mixture (4.4.50).

Treat seed with Thiram 5g

[Image not found: /uploads/2. Wheat Maize Barley Jute Cotton Sugarcane Groundnut_s114_img1.png]

Avoid close planting

Spray Benlet / Bavistine @.2%

Proper fertilization

Spray at 15 day interval.

Cultural practice which favours plant growth

Weed control

<div id="3-rice-disease-core-notes"></div>

Lecture: 3. Rice Disease Core Notes

Course Document: 3. Rice

Page 1

Blast Brown Spot Stem Rot

Sheath Blight

Piricularia oryzae Drechslera oryzae/

Bipolaris oryzae

Sclerotium oryzae Rhizoctonia solani

Leaf Blast: Spindle or eye

shaped lesion, ash colour in

centre with dark brown margin.

Under favourable conditions,

lesion enlarge

Collar Blast: Collar blast

occurs when the pathogen

infects the collar that can kill

the entire leaf blade.

Node Blast: Node turns

blackish and break easily

Panicle Blast: At the base of

panicle brown to black or rings

are formed.

Early Stage: The grains do not

fill as the food materials cannot

reach to the young grains and

ear turns white which is known

as white head symptoms.

Later Stage: The grains are

filled up partially, the rotten

neck cannot hold them and ear

breaks down at the point of

infection.

Attacks rice plant at all stages of its

growth.

Symptoms of the disease are spots on

the leaves and the glumes may also

appear on the leaf sheath, panicles

and stems.

The spots are more or less round,

dark brown colour in the centre

which is surrounded by yellow halo.

Culm and leaf sheath are affected

at water level.

These areas turn black and

become large.

Black powdery sclerotia are found

within the Culm and leaf sheath.

Spots or lesion near water level

Lesions are oval or elliptical

shaped

The centre of the lesion is light

coloured but margin dark

coloured.

Brownish sclerotia formed on the

lesion look like skin of cobra

snake.

Symptoms of Diseases of Rice

Page 2

Bakanae/Foot Rot Bacterial Blight

Ufra

Tungro

Fusarium moniliforme/

Gibberella fujikuroi

Xanthomonas campestris pv.

oryzae

Ditylenchus angustus Rice spherical tungro virus/ Rice

bacilliform tungro virus

Infected plants are several inches

taller than normal plants.

Plants are lin, thin and yellow

green in colour

Plants produce adventitious roots

at the lower node

Diseased plants bear few tillers

and the affected tillers die before

reaching maturity.

Lesion may develops from

anywhere on the leaf, but

generally they start at the edges

near the tip.

The edges of the lesions are

irregular and wavy.

Straw colour discoloration.

Chlorosis of the uppermost leaves

of rice plants which are more than

two months old.

Flag leaf and panicle can be

twisted and deformed

When the panicle remains largely

enclosed inside the boot is called

Thor ufra and when the panicle

emerges partially from the boot is

called Pucca ufra.

Plants become stunted

Number of tillers is reduced and

the leaves become yellow.

Yellowing ranges from light

yellow to orange yellow or

brownish yellow.

Pale yellow stripes are found

Stripes are parallel to the veins

Infected plants bear a few poor

panicles.

Page 3

Blast Brown Spot Stem Rot

Sheath Blight

*High relative humidity (90%

or higher)

*High nitrogen and low silica

content of the soil.

*Long dew period (10 hours or

more)

*Light rainfall or drizzle and

overcast sky.

*Close planting

*Soil with low nitrogen content.

*Deficiency of K, Mg and Silica

*Dry soil, too high or too low water

temperature.

*High relative humidity, continued

rain and cloudiness.

• Seedling infection takes place

when soil temperature below 25°C

*Stagnant water in the field

*Excessive nitrogen & low potash

fertilizer.

*Infected with stem borer.

*High humidity & warm

temperature.

*High nitrogenous fertilizer

*Close planting

Bakanae/Foot Rot Bacterial Blight

Ufra Tungro

*Soil temperature of 35°C

*Damp soil condition

*High nitrogenous fertilizer

application

*High temperature

*High nitrogenous fertilizer

*K & P Deficiency

*Excess silicon and magnesium

*Heavy rain flooding

Stagnancy of water and severe

wind.

*Temperature 28-30°C &

saturated RH (75%)

*Wettest areas & fields with high

water level

*Ufra infected rice is more

susceptible to the blast pathogen

*Presence of virus sources

*Presence of the vector.

*Age and susceptibility of the

host plant

*All growth stages of the rice

plant specially the vegetative

stage.

Suitable Factors of Diseases of Rice

Page 4

Blast Brown Spot Stem Rot

Sheath Blight

- Resistant varieties: BR-3, BR-4, BR-6,

BR-8, IR-22 etc

*Clean seed should be sown

*Seedling should be raised under low land

condition

*Avoid excess nitrogenous fertilizer

*Seed treatment with Granosan-M or

Agrosan-GN at the rate of 2 tolas/mound or

Vitavex-200 at the rate of 0.25% of seed

weight.

*Fungicides such as

Benlate/Topsinmethyl/Blasticidine/Hinosan

at the rate of 0.2% should be sprayed on the

leaves at the time of panicle emergence

*Cultivate resistant varieties:

Biplob, Brrisail, Progoti,

Mukta etc.

*Sowing healthy seeds

*Burning crop residues

*Hot water treatment (50°C

for 30 min)

*Balance fertilizer

Seedbed and land soil should

be kept wetty

- Benlate/ Hinosan at the rate

of 0.2% at 10-15 days interval

from 60 days of seedling.

*Resistant varieties: Biplob,

Borishal, Progoti, Mukta etc

should be cultivated.

*Stubble and residues should

be burnt.

*Crop rotation

*Drained out the water and

allowing soil to dry and crack

before re-watering

*Avoid excess nitrogen,

balance with K

*Spray with

Benlate/Homai/Topsin-M at

the booting and before

flowering stage at the rate of

0.2% at 9-15 days interval

The disease is hard to control-

*Varieties with moderate level

of resistance such as BR-4, BR-

5, IR-25, BAU-63 etc.

*Balance fertilizer and spacing

*Destroy crop residues

*Dry the field and re-water

*Spray with

Benlate/Hinosan/Topsin at the

rate of 0.2% at 10-15 days

interval

Bakanae/Foot Rot Bacterial Blight

Ufra Tungro

*Resistant varieties: Biplob,

Brrisall, Progati, Mukta etc

*Collection of seed from disease

free area

*Destroy crop residue

*Uprooting and destroying of the

infected plants

*Balance fertilizer should be used

*Seed treatment with Vitavax-200

*Resistant varieties such as IR-20,

IR-22, BR-4, BR-14, BAU-63 etc

*Balance fertilizer should be used

*Field should be dried and allow

to crack

*Plant residues should be burnt

*Seed should be collected from

healthy plant

*Resistant varieties such as Digha

*Burning the crop residues

*Summer ploughing

*Destruction of infected seedling

*Crop rotation

*Spreading of Furadan at the rate

of 5kg/bigha

*Seed treatment with Carbofuran

at the rate of 0.3kg/80kg

*Resistant varieties such as Mala,

Brisail, Kalojira, BR-5, 8, 10, 12

etc

*Weeds and infected plants

should be destroyed

*Application of insecticides to

control insect (leaf hopper)

Control of Diseases of Rice

Page 5

<div id="4-barley-maize-cotton-and-jute-core-notes"></div>

Lecture: 4. Barley, Maize, Cotton, and Jute Core Notes

Course Document: 4. Barley,Maize, Cotton and Jute

Page 1

Covered Smut of Barley Leaf Blight of Maize Angular Leaf Spot of Cotton

Ustilago hordei Drechslera turcicum Xanthomonas malvacearum

Symptoms develop during ears emerge

Every spikelet transformed into a sorus

At first the sori are covered with thick

membrane, which resists easy rupturing

The kernels are replaced by black spore

masses

Spores released at the time of threshing and

get attached to the seed.

Infected plants short

Yellowish, round or elliptical spots on leaf

Centre of the lesion brownish, black or dusty

Edges of the lesion is well defined

Spot is 5 inches length and 1 inch width

All parts are infected, specially leaf

Infected area become tan to brown

Tissues area become thin and semitransparent

Lesions first appear as small water soaked

area

Enlarged and turn brown to black with age

On cotyledons, round to irregular lesions

appear

Lesions develop along leaf veins and petioles

Lesions are limited by veinletes

Infected leaves turn yellow and fall to the

ground.

Symptoms

Page 2

Covered Smut of Barley Leaf Blight of Maize Angular Leaf Spot of Cotton

*When growth and development of plant is

good.

*Moist condition

*Factors favour to the crop

*Frequent rain

*In arid region, furrow irrigation

Stem Rot of Jute Black band of Jute Anthracnose of Jute

Macrophomina phaseolina Botryodiplodia theobromae Colletotricum corchorum

Plants are infected at all stages of its growth

Damping off of seedlings

In mature plants, buff to black lesions along

the apex and margin, mid rib and petioles

Irregular rotting along the stem

Fibre exposed

Plants shed leaves and die ultimately

Severe cases- capsule and seeds are affected

Numerous pycnidia develop on diseased parts

Dense black band found around the stem at

about 2-3 feet above the ground

Regular margin

Plants loose all leaves and dry black stem

Pathogen produces minute spherical pycnidia

on the black stem

On rubbing hands up and down on the infected

stem, the finger becomes black.

Yellowish-brown, water soaked & depressed

spots on the stem

The colour of the spots may become dark

brown and finally black

In most cases, infects vascular bundles

The lesion develops cracks in the centre & the fibres are exposed, called cankerous

symptoms.

The pathogen also infects the pods producing

necrotic lesions.

Suitable Factors

Page 3

Stem Rot of Jute Black band of Jute Anthracnose of Jute

*Moist weather and warm temperature

(>30°C)

*Excess nitrogen and low potash

Same as Stem Rot of jute Same as Stem Rot of jute

Covered Smut of Barley Leaf Blight of Maize Angular Leaf Spot of Cotton

*Seed treatment with Organomercurials at the

rate of 2.5 gm/kg seeds before sowing

*Fungicides such as Vitavex-200 are also

effective

*Resistant varieties should be used

*Disease free seeds should be used

*Resistant varieties should be used

*Dead leaves, stubbles and weeds should be

destroyed

*Crop rotation

*Spray with Tilt at the rate of 0.04%

*Disease free seed

*Burning/decomposition of infected plants

*Crop rotation

*Cultivate resistant varieties

*Defoliation with chemicals to reduce late boll

infection

Control

Suitable Factors

Page 4

Stem Rot of Jute Black band of Jute Anthracnose of Jute

*Diseases free seed should be used

*Destroy crop residues

*Crop rotation

*Application of balance fertilizer

*Seed treatment with Vitavex-200/Homai at

the rate of 0.25% of seed weight

*Spray Dithane M-45 at 15 days interval

*Resistant varieties-

C. capsularies: D154, CVL-1, CVE-3, CC-45

C. olitorius: O-4, O-9897

Same as Stem Rot of Jute Same as Stem Rot of Jute.

*Rotation with C. olitorius

Control

<div id="5-wheat-and-pulse-diseases-core-notes"></div>

Lecture: 5. Wheat and Pulse Diseases Core Notes

Course Document: 5. Wheat and Pulse

Page 1

Leaf Rust of Wheat Black Rust of Wheat Bipolaris Leaf Blotch and Black

Point of Wheat

Loose Smut of Wheat

Puccinia recondita Puccinia graminis var .tritici. Bipolaris sorokiniana

Perfect stage :Cochliobolus sativus

Ustilago tritici

Small orange or brown pustules

on the leaves.

Pustules are scattered at random

on leaves, not in stripes.

A chlorotic halo surrounds

individual pustules.

In the later stage, dark telia may

develop.

Generally appear at the latter

stages of plants development.

At first, elongated brown pustules

on the stem, leaf sheath and leaves

are appeared

Pustules may be a quarter of an

inch or more

Soon they burst and expose brown

uredospores.

At maturity, the pustules produce

mahogany-brown teliospores.

Plants are attacked at all stages of

growth

Causes post-emergence rot in

seedling

The symptoms in the seedling stage

appear as yellowish oval to oblong

spot

The spot gradually increase in size

and turn dark brown and spread over

the entire seedling.

In mature, straw colored patches are

produced.

Symptoms are conspicuous at

heading time.

Diseased ear emerge earlier

than healthy once

Entire panicle is replaced by a

mass of powdery, black

chlamydospores.

Page 2

Cercospora Leaf Spot of

Pulse

Mosaic of Mungbean Botrytis Gray Mould of Gram

(seed borne)

Rust of Pulse

Cercospora canescens Mungbean Yellow Mosaic Virus. Botrytis cinerea Uromyces ciceris arietini

*At first, water soaked pin

headed spots appear.

*This spots turn brown to

reddish brown. Then it

gradually increase in size.(dia-

1.5cm)

*The appearance of spots

differs variety to variety.

*In some variety brown hallow

with white centred spots.

*In severe cases, plants losses

their leaves.

Plants are attacked at all stages of

growth

Bright yellow patches intercepted by

green areas

In severe cases, leaves become

completely yellow

Infected plant becomes shortened

Produce smaller and fewer flower

Shedding of flowers and leaves.

Leaves covered with spore mass.

Affected flowers turn into a

rotting mass.

Lesions on stem are 10-30 mm

long and girdle the stem fully.

Tender branches break off at the

point where the gray mould has

caused rotting.

Lack of pod setting

Small oval brown powdery lesion

on leaves

Pustules may appear on either side

of the leaves

The entire leaf area covered with

rust pustules

Pustules appear on the pods, floral

parts & stem also

At advance stages teleutosorus

appear on the leaves

Page 3

Cercospora Leaf Spot of

Pulse

Mosaic of Mungbean Botrytis Gray Mould of Gram

Rust of Pulse

- Foggy weather

*Late rain

*Water lodging

*Heavy irrigation or rain

*Close planting

*Excessive vegetative growth

*Infected plant debris

*Within plant debris, it may

remain infective up to 8 months

*Low temperature[20-22]

*High humidity

Leaf Rust of Wheat Black Rust of Wheat Bipolaris Leaf Blotch and Black

Point of Wheat

Loose Smut of Wheat

*Dry windy days.

*Cool night with dew.

*Temperature 18-20°C

*Wind & mild humid weather

*A film of water on the leaf

surface is essential for

germination & penetration

*The presence of large acreage of

susceptible wheat

*Susceptible host

*Relative humidity above 90%

*Temperature 250 ± 20 C

*Poor nutrition

*Cold weather with light

showers.

*Dew during pollination time.

Cercospora Leaf Spot of

Pulse

Mosaic of Mungbean Botrytis Gray Mould of Gram

Rust of Pulse

Page 4

- Destroy crop residues

*Use of resistance variety

- Destroy weed hosts

*Crop rotation (soil borne)

*Optimum amount of fertilizer

• Spray with Bavistin @ 0.1%

at 7-10 days interval for 2-3

times

*Cultivation of resistant varieties:

BARI mung 5, 6/BARI mash1,2,3

*Spray with Malathion @ 0.2% to

control the insect vector.

*Uprooting and destroying of

infected plant (early stage)

• Destroy weed hosts

Cultural Control:

*Use of resistance variety

*Avoid excessive vegetative

growth

*Avoid excess irrigation

*Intercrop with linseed

*Deep summer ploughing

Biological Control:

*Use of Trichoderma spp.

Chemical Control:

*Seed treatment with

Thiram+Carbendazim (1:1) @

3g/kg

*Spray the crop with Captan @ 5-

6 kg/ha at 15 days interval

*Use of resistance variety

- Destroy weed hosts

*Seed treatment

*Chemical sprays & use of

sulphur dust

*Adjusting the planting time

Leaf Rust of Wheat Black Rust of Wheat Bipolaris Leaf Blotch and Black

Point of Wheat

Loose Smut of Wheat

- Destroy crop residues

*Use of resistance variety

*Cultivation resistant varieties eg.

High yielding Mexican varieties.

*Destruction of barberry plants.

- Resistant variety should be

cultivated.

*Seed should be collected from a

diseased free field.

*Burning of crop residues.

*Application of balanced fertilizer and

irrigation.

*Weed control.

*Crop rotation.

*Resistant varieties like

Kalyan 227

- Disease free seed.
- Hot water treatment at

54°C for 13 minutes.

*Seed treatment with

Vitavax-200 @ 0.25% of seed

weight.

<div id="6-sugarcane-mustard-sunflower-and-groundnut-core-notes"></div>

Lecture: 6. Sugarcane, Mustard, Sunflower, and Groundnut Core Notes

Course Document: 6. Sugarcane, Mustard, Sunflower, Groundnut

Page 1

Red Rot of Sugarcane Smut of Sugarcane Mosaic of Sugarcane Ratoon Stunting Disease (RSD)

Colletotrichum falcatum Ustilago scitaminea Sugarcane Mosaic Virus Clavibacter xyli

Recognized when canes are

nearing maturity.

Sugar content becomes low

Third to fourth leaf from top

turns paler and drops slightly.

The cane becomes light weight.

Dark reddish lesions develop

on the midrib of the leaf.

Discontinuous white/red

lesions appear when cross

sected

Cane becomes shrinkage

Long apex, whip-like, dusty, black,

shoot several feet in length and

curved

Abnormal growth of shoot, devoid of

leaves, slender and flexible

At initial stages, the shoot is covered

by a silvery-white thin membrane

Appear about 6 weeks after

planting & on maturity

Oval to elongated pale yellow

patches of irregular size on leaf

Appears mainly on young unfold

leaves

Also appears on stem

Growth is stunted.

Germination is reduced and

delayed.

Internodes become short.

Leaves yellowing

Cane becomes thin

Growth is affected during

summer, wilting

Pink colour near the nodes during

immature stage

Symptoms

Page 2

Grey Blight/Alternaria Leaf

Spot/Blight of Mustard

Leaf Spots of Sunflower

Leaf Spot of Groundnut (Tikka)

Alternaria brassicae Alternaria helianthi Cercospora arachidicola (Early

stage)

Cercosporidium personatum (Late

stage)

At first pin head spot appears

which increases later

Pale to brownish circular lesion

with dark concentric rings on

leaves

Spots also appear on stem & pod

Pods become sunken

Concentric ring with yellow

hallow

Spot irregular and circular shaped

In severe, whole leaf is destroyed

Cercospora arachidicola:

Symptoms first appear at the age of

1-2 months

In early spots, spores are formed

mostly on upper leaf surface

Brown red irregular circular spot

with yellow hallow.

Cercosporidium personatum:

Symptoms appear on both side of the

leaf during the later part of the

growing season

In later, spots, spores are formed

mostly on lower leaf surface

Spots are more circular and darker

than early spot

Page 3

Red Rot of Sugarcane Smut of Sugarcane Mosaic of Sugarcane Ratoon Stunting Disease (RSD)

- Germination of the buds in

unfavorable temperature and

moisture condition.

- Presence of stalk borer, moth

borer and sugarcane weevil.

• Injury to sett cutting.

Grey Blight/Alternaria Leaf

Spot/Blight of Mustard

Leaf Spots of Sunflower

Leaf Spot of Groundnut (Tikka)

*Foggy & Moist Weather are

harmful

*Long due period.

*Light rainfall& drizzle favorable.

*Excess N2 fertilizer.

*High moisture (above 95%)

*High temperature (above 25°C)

*Close planting

Suitable Factors

Page 4

Red Rot of Sugarcane Smut of Sugarcane Mosaic of Sugarcane Ratoon Stunting Disease (RSD)

*Use of resistant varieties e.g-

Iswardi- 2/54, Iswardi-16,

Iswardi-17etc.

*Setts should be collected from

disease free plant

*Dipping the cut end in coal-tar before planting.

*Infected plants should be

burnt

*Application of insecticide to

control borer

*Hot water treatment of setts

(55°c for 10minutes).

*Use of resistance variety

*Uprooting and destroy of diseased

plants

*Setts should be collected from

healthy plants

*Hot water treatment of setts (54°C

for 4 hours)

- Destroy weed hosts

*Use disease free sets

*Use of resistance variety

*Control insect vector timely

- Destroy infected cane
- Destroy weed hosts

*Use of resistance variety

- Use disease free sets
- Infected plants and roots should

be destroyed

*Avoid ratooning in the infected

field

*Hot water treatment of setts

(54°C for 4 hours)

Grey Blight/Alternaria Leaf

Spot/Blight of Mustard

Leaf Spots of Sunflower

Leaf Spot of Groundnut (Tikka)

*Destroy crop residues

*Rotation with non-crucifer crops

*Spray with Malathion @ of 0.2%

to control aphid

*Seed treatment with Vitavex-

200@ 0.25% of seed weight

*Early sowing before 25

November

*Plant in a area that receives early

morning sun

- Seed should be collected from

healthy plants

- Destroy crop residues

*Seed treatment with hot water &

chemical fungicide

*Spray 0.2% Dithane M-45

*Avoid close planting

*Proper fertilizer application & weed

control

*Crop residues should be burnt

• Spray with Bavistin @ 0.1% from

the first appearance of disease at 15-

20 days interval for 3-4 times

*Seed treatment with Thiram@ 5g/kg

seed

Control

<div id="7-pathogen-scientific-names-list"></div>

Lecture: 7. Pathogen Scientific Names List

Course Document: 7. Scientific Names

Page 1

Rice

Barley

Diseases Name Causal Organisms Medium

Covered Smut of Barley Ustilago hordei Seed borne

Leaf Rust Puccinia hordei

Stem Black Rust Puccinia graminis

Diseases Name Causal Organisms Medium

Fungal Diseases:

Blast Piricularia oryzae Seed borne

Brown Spot Drechslera oryzae/ Bipolaris oryzae Seed borne

Narrow brown Spot Cercospora oryzae Seed borne

Stem Rot Sclerotium oryzae Soil borne

Sheath Blight Rhizoctonia solani Soil borne

Bakanae/Foot Rot Fusarium moniliforme/ Gibberella

fujikuroi

Seed borne

Leaf Scald Gerlachia oryzae Seed borne

False Smut Ustilaginoidea virens

Sheath Rot Sarocladium oryzae

Grain Discoloration Bipolaris oryzae/Fusarium

miniliforme/Sarocladium

oryzae/Alternaria padwickii/Phoma

sorghina

Seed borne

Bacterial Diseases:

Bacterial Blight Xanthomonas campestris pv. oryzae

Bacterial Leaf Streak Xanthomonas campestris pv.

oryzicola

Nematode Diseases:

Ufra Ditylenchus angustus

White Tip Aphelenchoides besseyi

Root Knot Meloidogyne graminicola Soil borne

Virus Diseases:

Rice Tungro Disease

Rice tungro spherical virus (RTSV)

Rice tungro bacilliform virus (RTBV)

Rice Grassy Stunt Rice grassy stunt virus (RGSV)

Rice Hoja Blanca Rice hoja blanca virus (RHBV)

Rice Yellow Dwarf RYDV

Page 2

Maize

Diseases Name Causal Organisms Medium

Leaf Blight of Maize Drechslera turcicum

Anthracnose Colleotrichum graminicola

Maize Dwarf Mosaic Maize Dwarf Mosaic Virus

Cotton

Diseases Name Causal Organisms Medium

Angular Leaf Spot of Cotton Xanthomonas malvacearum Seed borne

Jute

Diseases Name Causal Organisms Medium

Stem Rot of Jute Macrophomina phaseolina Soil borne

Black band of Jute Botryodiplodia theobromae Soil borne

Anthracnose of Jute Colletotricum corchorum

Jute Mosaic Disease Jute Mosaic Virus (JMV)

Root Knot of Jute Meloidogyne javanica

Sugarcane

Diseases Name Causal Organisms Medium

Red Rot of Sugarcane Colletotrichum falcatum Soil borne

Smut of Sugarcane Ustilago scitaminea

Mosaic of Sugarcane Sugarcane Mosaic Virus

Ratoon Stunting Disease

(RSD)

Clavibacter xyli

Banded sclerotial Rhizoctonia solani Soil borne

Grassy Shoot of Sugarcane Mycoplasma(MLO's)

Oil

Diseases Name Causal Organisms Medium

Grey Blight/Alternaria Leaf

Spot/Blight of Mustard

Alternaria brassicae

Leaf Spots of Sunflower Alternaria helianthi

Leaf Spot of Groundnut (

Tikka)

Cercospora arachidicola

(Early stage)

Cercosporidium personatum

(Late stage)

Pulse

Diseases Name Causal Organisms Medium

Cercospora Leaf Spot of Pulse Cercospora canescens

Mosaic of Mungbean Mungbean Yellow Mosaic Virus

Botrytis Gray Mould of Gram

(seed borne)

Botrytis cinerea Seed borne

Rust of Pulse Uromyces ciceris arietini

Page 3

Wheat

Diseases Name Causal Organisms Medium

Leaf Rust of Wheat Puccinia recondita

Black Rust of Wheat Puccinia graminis var .tritici.

Bipolaris Leaf Blotch and

Black Point of Wheat

Bipolaris sorokiniana

Perfect stage :Cochliobolus

sativus

Loose Smut of Wheat Ustilago tritici Seed borne

<div id="8-comparative-analysis-disease-differences"></div>

Lecture: 8. Comparative Analysis & Disease Differences

Course Document: 8. All Disease Differences

Page 1

Blast of Rice Brown Spot of Rice

Lesion eye or spindle shaped Spot more or less round

Centre ash coloured Centre dark brown coloured

Margin of the lesion is dark brow coloured Margin of the spot is yellow coloured

Rich man disease Poor man disease

Lesion may increase Spot is localized

Tungro N2 Deficiency

Scattered yellowing in the field Yellowing in the whole field

No change of colour after application of urea change of color after application of urea

When iodine is mixed with infected leaves,

colour of iodine changed

When iodine is mixed with deficient leaves,

colour of iodine changed

Stripe may be found Stripe is not found

Stem Rot of Rice Sheath Blight of Rice

Lesions are not oval or elliptical Lesions are oval or elliptical

Lesion does not look like skin of cobra snake Lesion looks like skin of cobra snake

Black powdery sclerotia are found Brownish sclerotia are found

Resistant varieties are- Biplob, Borishal,

Progoti Mukta etc

Resistant varieties are- BR-4, BR-5, IR-26,

BAU-63 etc

Loose Smut of Barley Covered Smut of Barley

Ear emerges early Ear emerges later

Ear is replaced by mass of powdery black

chlamydospore

Ear is replaced by black smut sori

Covered by silver membrane Covered by gray membrane

Ustilago tritici Ustilago hordei

Stem Rot of Jute Black Band of Jute

Margin of the lesion is irregular Margin of the lesion is regular

Plant loose leaves and remain in the field as

black stem

Plant shed leaves and stem become rot,

ultimately plant will die

On rubbing hand ups and down on infected

stem, the figure doesn't become black

On rubbing hand ups and down on infected

stem, the figure become black

Pycnidiospores are single celled Pycnidiospores are double celled

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Lecture: 9. Core Disease Cycles

Course Document: All Disease Cycles

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Lecture: 10. Disease Visual Identification Guide

Course Document: Disease Picture

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Lecture: 11. Symptoms of Major Crop Diseases

Course Document: Symptoms of Important Diseases

Page 1

BLAST OF RICE BROWN SPOT OF RICE

STEM ROT OF RICE SHEATH BLIGHT OF

RICE

Collar blast

Collar blast occurs when the pathogen infects the collar that can kill the entire leaf blade.

Node blast

The pathogen also infects the node of the stem that turns blackish and breaks easily; this condition is called Node blast.

Leaf:

Spindle or eye shaped lesion, ash colour in center with dar k brown margin.

Under favorable conditions, lesions enlarge and coalesce eventually killing the leaves.

Panicle:

At the base of panicle brown to black spots or rings are formed called – “Neck Blast/ Panicle Blast”.

Early neck infection:

The grains do not fill as the food

materials can

not reach to the young grains

and ear turns

white which is known as white

head symptoms.

At later stage neck infection:

The grains are filled up partially

or incompletely, the rotten neck

can not hold them and ear breaks

down at the point of infection

Attacks rice plant at all stages

of its growth.

On the coleopt ile the spots

are brown and circular to

oval.

One or more spots may

coalesce to form bigger

discolored zones.

Symptoms of the disease are

spots on the leaves and the

glumes, may also appear on

the leaf sheath, panicles

and stems.

Spots on the leafs first appear

as minute deep brown dots.

The spots enlarge and

become circular to oval and

dark brown.

The spots are surrounded by

a yellow halo.

When glumes are infected

black spots appears on the

surface.

In severe cases the whole

surface of the glumes may be

coated with dark brown

mycelium and spores and

form a velvety mat.

Irregular water soaked areas

are appeared on sheath at

water line

These areas turn black and

become large.

Excessive production of the

late tillers and rotten in severe

cases

Black powdery sclerotia are

found within the culm and

leaf sheath.

Spots or

lesion near

water level

Lesions are

oval or

elliptical

Mature

lesions are grayish-white

Margin is dark coloured

Brownish sclerotia form on

lesion

Look like sk in of cobra

snake

BACTERIAL LEAF

STREAK OF RICE

NARROW BROWN

LEAF SPOT OF RICE

Initially water -

soaked fine,

interveinal,

long/ short

streaks/lesions

found on the

leaves

Lesions become

enlarge

gradually, borders turn

brownish and the centre

remain gray.

On lesions - minute yellowish

beads of dried bacterial

exudates may be found.

A large portion of leaf blade

becomes yellow which is

confused with tungro.

symptoms usually observed

during the later growth

stages.

short, linear, brown lesions

found on the leaves and also

on leaf sheaths, pedicels,

and glumes.

The lesions are 2 to 10 mm

long and 1 mm wide

A net blotch - like pattern

forms on leaf sheaths

TUNGRO OF RICE

UFRA DISEASE OF RICE

The first symp toms are

cholorosis with white patches

or striping of the uppermost

leaves of rice plants which are

more than two months old.

The leaves including flag leaf

and panicle can be twisted and

deformed with a frequently

wavy margin.

In severe attack, two

distinct types of symptoms are

noted on the panicle –

when the panicle remain s

largely enclosed in side the

boot, while the stalk partly

emerges and shows branching,

it is called Swollen or Kutcha

or Thor ufra.

when the panicle emerges it

gets distorted with a dark

brown discolored peduncle

and produces normal grains

only near the tip but the grains

below are sterile and

shrivelled. it is called Ripe or

pucca ufra.

The plants become stunted.

Number of tillers are

reduced and the leaves

become yellow.

Tillering is influenced by

the age of the infected

plants.

Yellowing ranges from light

yellow to orange yellow or

brownish yellow starts from

the tip of the leaves.

Pale yellow stripes are

found.

The

stripes

are

parallel

to the

veins.

The

yellow orange color persists

for sometimes and the

leaves die.

Infected p lants bear a few

poor panicles.

BAKANAE/ FOOT ROT

DISEASE OF RICE

BACTERIAL LEAF

BLIGHT OF RICE

Infected plants are several

inches taller than normal plants

in seedbed and field

Plants are slender, thin and

yellowish green in color

Plants produce adventitious

roots at the lower node of the

culms.

Diseased plants bear few tillers

and leaves dry up quickly. The

affected tillers usual ly die

before reaching maturity.

Partially filled grains, sterile, or

empty grains for surviving plant

at maturity

Symptoms appear on the

leaves of young plants as

pale-green to grey-green,

water-soaked streaks near the

leaf tip and margins.

These lesions coalesce and

become yellowish-white with

wavy edges.

The whole leaf may

eventually be affected,

becoming whitish or greyish

and then dying. Systemic

infection results in wilting,

desiccation of leaves and

death, particularly of young

transplanted plants.

In older plants, the leaves

become yellow and then die.

Symptoms

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Page 2

STEM ROT OF JUTE BLACK BAND OF JUTE ANTHRACNOSE OF

JUTE

COVERED SMUT OF

BARLEY

Plants are infected at all stages

of growth.

Damping off the seedlings

In minute plants, buff to black

lesions

along the

apex and

margin,

mid rib

and

petioles.

Irregular

rotting along the stem.

Fibre exposed, plants shed

leaves and die ultimately.

Severe cases -capsule and

seeds are affected.

Numerous pycnidia develop

on diseased parts.

Formation of dense black band

around the stem at about 2 -3

feet above the ground

Regular margin

Affected plants

ultimately loose

all the leaves and

remain as dry

black stems in

the field

The bark of the

main stem splits longitudinally

at later stages of disease

development

Fiber turns brown and dry.

Pathogen produces minute

spherical black pycnidia on the

blackened stem.

On rubbing hands up and down

on the infected stem, the finger

becomes black due to blackish

pycnidiosores formed by

pycnidia.

Yellowish-brown water

soaked & depressed spots

on the stem.

The colour of the spots may

become dark brown and

finally black.

In most cases, infects

vascular bundles. The lesi on

develops cracks in the centre

and the fibres are ex posed,

called cankerous symptoms.

The pathogen also infects the

pods producing necrotic

lesions.

Symptoms

develop when

ears emerge

out.

Every spikelet

transformed

into a sorus.

At first sori

are covered

with thick

membrane, which resists easy

rupturing.

The kernel are replaced by

black spore masses.

Spores released at the time of

threshing and get attached to

the seed.

Infected plants become short.

LEAF BLIGHT OF

MAIZE

ANGULAR LEAF SPOT

OF COTTON

LEAF RUST OF WHEAT BLACK RUST OF

WHEAT

Yellowish, round or elliptical

spots on leaf, centre of the

lesion brownish, black or

dusty.

Edges of the lesion is well

defined, spot is 5 inches

length and 1 inch width. All

parts are infected, specially

leaf.

Infected

area

become tan

to brown.

Tissues area

become thin and

semitransparent.

Lesions 1 st appear as small

water soaked area, enlarges

and turn brown to black with

age. On co tyledons, round to

irregular lesions appear.

Lesions develop along leaf

veins and petioles, lesion s are

limited by veinletes. Infected

leaves turn yellow and fall to

the

ground.

Small, circular, orange -

brown pustules on the leaves.

Pustules are scattered at

random over leaves, not in

stripes.

A chlorotic halo surrounds

individual pustules.

It appears on leaf sheath,

penduncles, internodes and

ear heads.

In the later stage, dark telia

may

develop.

Generally appear at the altter

stages of plants development.

At first, elongated brown

postules on the stem, leaf

sheath and leaves are

appeared.

Pustules appear gradual

change from brown to black.

The epidermis ruptures and

expose a black bed of

teliospore.

At maturity, the pustules

produce mahogany-brown

teliospores.

also expose brown

urediospores

BIPOLARIS SPOT BLOTCH &

BLACK POINT OF WHEAT LOOSE SMUT OF WHEAT MOSAIC OF JUTE ROOT KNOT OF JUTE

Plants are attacked at all

stages of

growth,

causes post

emergence

rot in

seedling.

The

symptoms in

the seedling stage appear.

Appears on leaf,

sheath,node and glumes

Light brown lesions

mostly oval to oblong

shape

Dark brown margin with

light brown center

Symptoms are conspicuous at

heading time.

Head of affected plants

converted into black mass of

spores &

no grain

formation

Affected

young

spike lets

covered

by silvery

membrane

Affected young spike lets

covered by silvery membrane

The spores are released by

wind or rain & only the central

axis is left at harvest.

Yellow flakes gradually

increase in size to form green

and chlorotic intermingled

patches producing a yellow

mosaic appearance

Stunted growth of the plant.

Defoliation at severe

infestation.

Root-knot nematodes do not

produce any specific above-

ground symptoms. Affected

plants have an unthrifty

appearance and often show

symptoms of stunting, wilting

or chlorosis (yellowing).

Symptoms are particularly

severe when plants are

infected soon after planting.

Below ground, the symptoms

of root -knot nematodes are

quite distinctive. Lumps or

galls ranging in size from 1 to

10 mm in diameter, develop

all over the roots. In severe

infestations, heavily galled

roots may rot away, leaving a

poor root system with a few

large galls.

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Lecture: 12. MRI Lecture Notes

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Lecture: 13. MRI Book Excerpts & Study References

Course Document: MRI Book

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Lecture: 14. Consolidated Pathology Study Guide

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Lecture: 15. Rezaul Sir Final Review Sheets

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Revision Questions & Detailed Answers

Rice Diseases

Q1: List six major diseases of rice with causal agents. [Paper 1]

Source: Original Course Content

Answer

Here are six major diseases of rice along with their causal agents and taxonomic groups:

#	Disease Name	Causal Agent (Scientific Name)	Pathogen Type
1	**Rice Blast**	*Pyricularia oryzae* (syn. *Magnaporthe oryzae*)	Fungus
2	**Brown Spot of Rice**	*Bipolaris oryzae* (syn. *Drechslera oryzae*)	Fungus
3	**Bacterial Leaf Blight (BLB)**	*Xanthomonas oryzae* pv. *oryzae*	Bacteria
4	**Sheath Blight**	*Rhizoctonia solani*	Fungus
5	**Bakanae / Foot Rot**	*Fusarium moniliforme* (syn. *Gibberella fujikuroi*)	Fungus
6	**Ufra Disease**	*Ditylenchus angustus*	Nematode
7	**Rice Tungro Disease**	*Rice tungro spherical virus* (RTSV) & *Rice tungro bacilliform virus* (RTBV)	Virus

Source: Lecture notes — 7. Scientific Names, Page 1

Q2: Draw and describe the symptoms of bacterial leaf blight and brown spot of rice. [Paper 1, 5, 6]

Source: Original Course Content

Answer

1. Bacterial Leaf Blight (BLB) of Rice

- **Causal Agent:** *Xanthomonas oryzae* pv. *oryzae* (formerly *Xanthomonas campestris* pv. *oryzae*)
- **Symptoms:**
- **Leaf Lesions:** Symptoms usually start near the leaf tips or margins as water-soaked stripes. These stripes enlarge, turn pale yellow to straw-colored, and eventually turn dry and greyish-white.
- **Wavy Margins:** The edges of the active lesions are distinct, characteristically **irregular and wavy**.
- **Bacterial Ooze:** In humid conditions, milky or turbid droplets of bacterial ooze (exudate) may form on young lesions, drying up into tiny amber-colored crusts.
- **Cresek Stage (Systemic Infection):** If infection occurs in young seedlings (tillering stage), it leads to systemic wilting where leaves roll, wither, and the entire seedling dies.

2. Brown Spot of Rice

- **Causal Agent:** *Bipolaris oryzae* (syn. *Drechslera oryzae* / *Helminthosporium oryzae*)
- **Symptoms:**
- **Leaf Spots:** Numerous small, circular to oval spots (resembling sesame seeds) appear on the leaves.
- **Coloration:** The center of the spots is **dark brown to light brown**, surrounded by a distinct **light yellow halo**.
- **Plant Stages:** Attacks the rice crop at all growth stages (seedling to maturity).
- **Glume Infection:** Also attacks glumes, producing dark brown to black spots (grain discoloration), which severely reduces seed viability and quality (historically led to the **Bengal Famine of 1943**).

Source: Lecture notes — 3. Rice, Pages 1-2; 8. All Disease Differences, Page 1

Q3: Illustrate and describe the disease cycle of brown spot of rice. [Paper 1, 5, 6, 8]

Source: Original Course Content

Answer

Disease Cycle of Brown Spot of Rice (Bipolaris oryzae)

The disease cycle describes the overwintering, primary infection, dissemination, and secondary cycles of the fungus.

1. Overwintering (Survival)

- **Primary Inoculum:** The pathogen overwinter/survives as **dormant mycelium** or **conidia** inside or on the seed (seed-borne) and in infected crop residues/stubbles in the field (soil-borne).
- **Collateral Hosts:** It can also survive on wild grasses and weeds surrounding the rice fields.

2. Primary Infection

- When contaminated seeds are sown, germinating mycelium infects the coleoptile and young leaves, causing seedling blight and rot.
- Sclerotia or mycelium in plant debris germinate in warm, moist conditions to infect low-lying foliage.

3. Secondary Infection (Dissemination)

- **Conidia Production:** The primary lesions produce numerous conidiophores that bear conidia (spores) in succession.
- **Dissemination:** The conidia are easily detached and disseminated by wind, light rain, and air currents to healthy foliage.
- **Secondary Cycle:** Under high humidity (>90% RH) and warm temperatures (25-30°C), conidia germinate, penetrate the host tissue through stomata or directly, and produce new spots within 6-8 days. This repeating cycle continues throughout the growing season.

4. Grain Infection

- During flowering and milk stages, wind-borne conidia land on the glumes, infecting the grain coat and embedding mycelium, completing the cycle.

```
* graph TD * A[Infected Seeds / Crop Residues] -->|Primary Infection| B[Seedling Germination & Blight] * B -->|Mycelial Invasion| C[Leaf Lesions / Sesame spots on Leaves] * C -->|Conidiophores produce Conidia| D[Conidia Disseminated by Wind & Rain] * D -->|Secondary Infection| E[Healthy Rice Leaves Infected] * E --> C * C -->|Infection at Heading Stage| F[Glume / Grain Infection] * F -->|Harvesting| A
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Source: Lecture notes — 3. Rice, Pages 1, 3, 4

Q4: Describe the symptoms of blast disease at different growth stages of rice. [Paper 2, 8]

Source: Original Course Content

Answer

Rice Blast (Pyricularia oryzae) is highly destructive because it attacks the plant at almost all growth stages, presenting four distinct symptom phases:

1. Leaf Blast Stage (Vegetative Stage)

- **Lesions:** Typical spindle-shaped or **eye-shaped spots** (tapering at both ends) appear on the leaves.
- **Coloration:** The center of the spot is **grey or ash-colored**, while the margins are dark brown or reddish-brown.
- **Blighting:** In highly susceptible varieties, the leaf spots enlarge rapidly, coalesce, and cause the entire leaf to dry up and die, giving the field a burnt look.

2. Collar Blast Stage

- **Location:** The pathogen infects the junction of the leaf blade and the leaf sheath (the collar).
- **Effect:** A dark brown lesion forms around the collar. This cuts off nutrient flow to the leaf blade, killing the entire leaf blade and causing it to drop off.

3. Node Blast Stage (Tillering/Flowering)

- **Location:** The pathogen attacks the nodal joints of the stem.
- **Symptoms:** The infected node turns black, dry, and necrotic.
- **Lodging:** The infected nodal tissue becomes weak and fragile, breaking easily under wind pressure or grain weight, causing the plants to fall over (lodge).

4. Neck Blast / Panicle Blast Stage (Heading Stage)

- **Location:** The pathogen infects the neck of the panicle (just below the grain head).
- **Symptoms:** The neck tissue turns brown to black, rots, and is covered with greyish mycelial growth.
- **Effects:**
- **Early Infection (Milk stage):** The neck rots before grains fill. The panicle remains erect, but the grains are completely empty and light-colored (**white head / chaffy grains**).
- **Late Infection (Dough stage):** Grains are partially filled, but the rotten neck breaks under the weight of the panicles, causing them to hang down or drop off.

Source: Lecture notes — 3. Rice, Pages 1-2

Q5: Analyze the favorable conditions of blast disease of rice. [Paper 2]

Source: Original Course Content

Answer

The development and epidemic spread of Rice Blast (Pyricularia oryzae) are highly dependent on specific environmental and agronomic factors:

1. High Relative Humidity

- Saturated atmospheric humidity (**90% or higher**) is extremely favorable. Free moisture or water droplets on the leaf surface are essential for conidial germination and appressorium formation.

2. High Nitrogenous Fertilization

- Excessive application of Nitrogen (N) fertilizer makes the host plant tissues succulent and soft, drastically reducing cell wall thickness and making them highly susceptible to fungal penetration.

3. Low Silica Content

- Inadequate silica (Si) in the soil weakens the physical barrier of the leaf epidermis, facilitating direct fungal penetration.

4. Wet Weather and Low Temperature

- Cool night temperatures (**25-28°C** or slightly lower) combined with a long dew period (**10 hours or more**), light drizzle, cloudy skies, and mist create perfect conditions for spore germination and colonization.

5. High Planting Density (Close Planting)

- High plant population creates a dense canopy that traps humidity, reduces sunlight penetration, and prolongs leaf wetness, accelerating pathogen multiplication.

Source: Lecture notes — 3. Rice, Page 3

Q6: Suggest the measures to be taken to grow blast disease free rice crop. [Paper 2, 5]

■ *Source: Original Course Content*

Answer

Growing a blast-free (Pyricularia oryzae) rice crop requires an Integrated Disease Management (IDM) strategy:

1. Cultivate Resistant Varieties

- Grow locally recommended blast-resistant varieties such as **BR-3, BR-4, BR-6, BR-8, IR-22**, etc.

2. Balanced Fertilization

- **Avoid excessive nitrogenous fertilizers.** Apply nitrogen in split doses.
- Apply adequate Potash (K) and Silica (Si) fertilizers, which strengthen plant cell walls and increase natural physical resistance.

3. Clean Cultivation & Sanitation

- Burn or destroy crop residues (straw and stubble) after harvest to eliminate overwintering mycelium.
- Keep fields and borders free of wild grass weeds, which act as collateral hosts.

4. Seed and Seedbed Management

- Raise seedlings under wetland conditions instead of dry seedbeds, as dry soil increases susceptibility.
- Sow healthy, clean seeds from certified disease-free fields.

5. Chemical / Seed Treatment

- **Seed Treatment:** Treat seeds with fungicides like **Vitavax-200** or Homai @ 0.25% of seed weight to eliminate seed-borne inoculum.
- **Foliar Spray:** Spray systemic fungicides such as **Tricyclazole (Trooper)** or **Hinosan / Edifenphos** @ 0.2% on leaves at the boot stage and panicle emergence if early leaf spots appear.

Source: Lecture notes — 3. Rice, Page 4

Q7: List two fungal, two bacterial, two nematode and two viral diseases of rice with their causal agents. [Paper 3, 6]

Source: Original Course Content

Answer

Here is a list of major rice diseases categorized by pathogen type, along with their causal agents:

1. Fungal Diseases

1. **Rice Blast:** *Pyricularia oryzae* (syn. *Magnaporthe oryzae*)
2. **Brown Spot of Rice:** *Bipolaris oryzae* (syn. *Drechslera oryzae*)

2. Bacterial Diseases

1. **Bacterial Leaf Blight (BLB):** *Xanthomonas oryzae* pv. *oryzae*
2. **Bacterial Leaf Streak (BLS):** *Xanthomonas oryzae* pv. *oryzicola* (syn. *Xanthomonas campestris* pv. *oryzicola*)

3. Nematode Diseases

1. **Ufra Disease:** *Ditylenchus angustus*
2. **White Tip Disease:** *Aphelenchoides besseyi*

4. Viral Diseases

1. **Rice Tungro Disease:** *Rice tungro spherical virus* (RTSV) & *Rice tungro bacilliform virus* (RTBV) (co-infection)
2. **Rice Grassy Stunt:** *Rice grassy stunt virus* (RGSV)

Source: Lecture notes — 7. Scientific Names, Page 1

Q8: Write down the symptoms of brown spot and sheath blight of rice. [Paper 3]

Source: Original Course Content

Answer

1. Symptoms of Brown Spot of Rice (Bipolaris oryzae)

- **Attacked Parts:** Attacks all stages of growth. Spots occur on leaves, leaf sheaths, glumes, and stems.
- **Lesions:** Spots are **more or less round or oval**, typical size of sesame seeds.
- **Coloration:** Center of the spot is dark brown surrounded by a **distinct yellow halo**.
- **Grains:** Maturing grains are discolored with dark brown spots on glumes.

2. Symptoms of Sheath Blight of Rice (Rhizoctonia solani)

- **Location:** Lesions first develop on the leaf sheaths near the water level.
- **Lesions:** Spots are large, **oval or elliptical**, greyish-green in color.
- **Center & Margin:** Center of the lesion turns straw-colored or off-white, while the margins are dark reddish-brown.
- **Cobra Skin Pattern:** The lesions enlarge, coalesce, and cover large areas of the sheath, resembling the **skin of a cobra snake**.
- **Sclerotia:** Under humid conditions, large, brownish-black globose sclerotia form loosely on the infected sheaths and drop into water.

Source: Lecture notes — 3. Rice, Page 1; 8. All Disease Differences, Page 1

Q9: How can you distinguish between brown spot and blast diseases of rice in the field? [Paper 3]

Source: Original Course Content

Answer

Rice Blast and Brown Spot can be distinguished in the field using the following criteria:

Feature	Rice Blast (*Pyricularia oryzae*)	Brown Spot (*Bipolaris oryzae*)
Lesion Shape	**Spindle-shaped or eye-shaped** (narrowing at both ends).	**Round or oval-shaped** (like sesame seeds).
Lesion Center	Light grey or **ash-colored** center.	**Dark brown** center.
Lesion Margin	Dark brown or reddish-brown border.	Surrounded by a **bright yellow halo**.
Lesion Growth	Enlarge rapidly, coalesce, blighting the entire leaf.	Spots remain localized and circular.
Socio-Economic Link	**"Rich man's disease"** (prefers high nitrogen, high fertility fields).	**"Poor man's disease"** (prefers nutrient-deficient, low nitrogen soils).
Crop Growth Stage	Most destructive at heading/flowering stage (neck blast).	Destructive at all stages, especially seedling and grain filling.

Source: Lecture notes — 8. All Disease Differences, Page 1

Q10: Enumerate the symptoms of blast and bacterial leaf blight of rice with diagram. [Paper 4]

■ *Source: Original Course Content*

Answer

1. Symptoms of Rice Blast (Pyricularia oryzae)

- **Leaf Phase:** Spindle-shaped or eye-shaped spots with ash-grey centers and dark brown margins.
- **Nodal Phase:** Nodal joints turn black, rot, and lodge (break).
- **Neck Phase:** Rotting at the base of the panicle (neck turns brown/black). Grains remain empty ("white heads") or neck breaks.

2. Symptoms of Bacterial Leaf Blight (BLB) (Xanthomonas oryzae pv. oryzae)

- **Leaf Lesions:** Starts as water-soaked lesions near the leaf tip or margins. Enlarges and turns straw-colored or greyish-white.
- **Margins:** Characterized by **irregular, wavy margins** along the lesion edge.
- **Cresek Stage:** Systemic infection leading to wilting of the whole seedling.
- **Exudate:** Milky bacterial ooze drops appear on the leaves in humid mornings, drying to amber crusts.

(For exam diagrams: Draw a spindle-shaped spot with ash center/brown border for blast; draw a leaf showing a straw-colored blighted tip with wavy, irregular side margins for BLB).

Source: Lecture notes — 3. Rice, Pages 1-2; 8. All Disease Differences, Page 1

Q11: Prescribe the control measures of stem rot and brown spot of rice. [Paper 4, 7]

Source: Original Course Content

Answer

Prescription for managing Stem Rot and Brown Spot of Rice:

1. Prescription for Stem Rot (Sclerotium oryzae)

- **Resistant Varieties:** Grow resistant cultivars like **Biplob, Borishal, Progoti, Mukta**, etc.
- **Sanitation:** Burn stubbles and crop residues after harvest to destroy soil-borne sclerotia.
- **Water Management:** Drain out stagnant water from infected fields. Allow the soil to dry and crack completely before re-watering.
- **Fertilization:** Avoid excess nitrogen. Maintain balanced fertilization with a high ratio of Potash (K).
- **Foliar Spray:** Spray **Benlate, Homai, or Topsin-M** @ 0.2% at the booting stage and repeat before flowering at 9-15 days interval.

2. Prescription for Brown Spot (Bipolaris oryzae)

- **Resistant Varieties:** Grow cultivars like **Biplob, Brrisail, Progoti, Mukta**, etc.
- **Seed Selection:** Sow clean, healthy seeds from disease-free fields.
- **Seed Treatment:** Perform **Hot Water Treatment** (soak seeds at 50°C for 30 minutes) or treat seeds with **Vitavax-200** @ 0.25% seed weight.
- **Fertilization:** Apply balanced fertilizers. Brown spot is a "poor man's disease", so ensure adequate Nitrogen (N), Potash (K), Magnesium (Mg), and Silica (Si) are applied to deficient soils.
- **Foliar Spray:** Spray **Benlate or Hinosan** @ 0.2% at 10-15 days interval starting from 60 days of seedling.

Source: Lecture notes — 3. Rice, Page 4

Q12: Describe the symptoms of brown spot and bacterial leaf blight of rice with labeled diagram. [Paper 5]

■ *Source: Original Course Content*

Answer

Refer to the comparative descriptions of Brown Spot and BLB:

1. Brown Spot (Bipolaris oryzae)

- Small, circular or oval dark brown spots (sesame seed size) on leaves.
- Spots have a light brown/greyish center and are surrounded by a **distinct yellow halo**.
- Spots are localized and appear at all growth stages.
- Maturing grains show dark brown spots on glumes.

2. Bacterial Leaf Blight (BLB) (Xanthomonas oryzae pv. oryzae)

- Yellowish, straw-colored lesions starting from the tip or edges of the leaf.
- The edges of the lesions are characteristically **irregular and wavy**.
- Systemic seedling wilting (**Cresek stage**).
- Droplets of milky bacterial ooze under humid conditions.

(Labeled diagram guide: Draw a leaf showing multiple small oval brown spots with yellow halos for brown spot. Draw a leaf with a blighted yellow-white margin extending from the tip downwards with a wavy boundary for BLB).

Source: Lecture notes — 3. Rice, Pages 1-2; 8. All Disease Differences, Page 1

Q13: How can you control blast and stem rot of rice? [Paper 5]

Source: Original Course Content

Answer

1. Control Measures for Rice Blast (Pyricularia oryzae)

- **Resistant Cultivars:** Use blast-resistant varieties such as **BR-3, BR-4, BR-6, BR-8, IR-22**, etc.
- **Agronomic Practices:** Avoid dry seedbeds. Avoid excess application of nitrogenous fertilizers. Split nitrogen applications. Apply potassium and silicon.
- **Sanitation:** Burn stubbles and crop residues to destroy overwintering inoculum.
- **Seed Treatment:** Treat seeds with **Vitavax-200** @ 0.25% of seed weight.
- **Foliar Fungicides:** Spray systemic fungicides like **Tricyclazole (Trooper)** or **Hinosan / Edifenphos** @ 0.2% at booting and panicle emergence.

2. Control Measures for Stem Rot (Sclerotium oryzae)

- **Resistant Cultivars:** Cultivate resistant varieties like **Biplob, Borishal, Progoti, Mukta**, etc.
- **Water Management:** Stem rot is favored by stagnant water. Drain fields, allow the soil to dry and crack before re-watering.
- **Fertilization:** Balanced fertilization (avoid excessive N, apply adequate K).
- **Sanitation:** Burn infected crop stubble and trash immediately after harvest.
- **Chemical Spray:** Spray foliar fungicides like **Benlate, Homai, or Topsin-M** @ 0.2% at the booting stage and before flowering.

Source: Lecture notes — 3. Rice, Page 4

Q14: Describe the symptoms of brown spot and false smut of rice with clear diagram. [Paper 6]

Source: Original Course Content

Answer

1. Symptoms of Brown Spot (Bipolaris oryzae)

- Localized round or oval spots (sesame seed size) on leaves, glumes, and leaf sheaths.
- Distinct dark brown center surrounded by a **yellow halo**.

2. Symptoms of False Smut of Rice (Ustilaginoidea virens)

- **Affected Parts:** The pathogen attacks only the individual grains of the panicle during the flowering stage.
- **Smut Balls:** Individual grains are transformed into **large, velvety, globose, orange-green or dark-green spore balls** (smut balls).
- **Scale:** The smut balls can grow to twice the size of a normal grain, enclosing the floral parts.
- **Color progression:** The spore ball is initially orange-yellow, covered by a membrane, which later ruptures, turning olive-green to chalky black.
- **Yield Loss:** Only a few grains per panicle are affected, but adjacent grains fail to fill due to pathogen toxins.

(Diagram guide: For brown spot, draw oval spots with concentric zones on leaf. For false smut, draw a rice panicle where 2-3 grains are replaced by large, puffy, globose velvety balls).

Source: Lecture notes — 3. Rice, Page 1; 7. Scientific Names, Page 1

Q15: What are the suitable factors of disease development of stem rot and bacterial leaf blight of rice? [Paper 6, 5]

■ *Source: Original Course Content*

Answer

1. Favorable Factors for Stem Rot (Sclerotium oryzae)

- **Stagnant Water:** Constant presence of stagnant irrigation water in the field allows sclerotia to float, germinate, and infect leaf sheaths at the water level.
- **High Nitrogenous Fertilizer:** High N fertilizer yields succulent host tissues; low Potash (K) weakens structural cell walls.
- **Insect Injury:** Infestation by **stem borers** creates wounds, facilitating pathogen entry into the culm.

2. Favorable Factors for Bacterial Leaf Blight (BLB) (Xanthomonas oryzae pv. oryzae)

- **High Temperature:** Warm temperatures between **25-30°C** are highly favorable.
- **High Nitrogenous Fertilizer:** Encourages rapid vegetative growth and soft succulent tissue.
- **Nutrient Imbalance:** Deficiency in Potash (K) and Phosphorus (P), and excess Silicon and Magnesium.
- **Heavy Rainfall & Flooding:** Heavy rains and flooding disseminate bacterial cells across fields.
- **Severe Winds:** Strong winds cause leaves to rub against each other, creating micro-wounds that allow bacterial entry.

Source: Lecture notes — 3. Rice, Page 3

Q16: Enumerate the importance of balance fertilizer application for rice disease control. [Paper 6]

Source: Original Course Content

Answer

Balanced fertilizer application is a core pillar of cultural disease management in rice. The ratio of macronutrients and micronutrients directly alters host physiology and structural resistance:

1. Nitrogen (N) — The Promoter of Susceptibility

- **Effect of Excess N:** Excessive application of Nitrogen makes plant tissues highly succulent and soft, reduces cell wall thickness, and increases the size of stomata.
- **Resulting Diseases:** Greatly increases susceptibility to **Rice Blast, Sheath Blight, Stem Rot, and Bacterial Leaf Blight (BLB)**.
- **Control:** Splitting N applications and applying recommended doses prevents succulent growth.

2. Potash (K) — The Disease Suppressor

- **Effect of K:** Potash strengthens cell walls, accelerates the synthesis of phenols, lignin, and phytoalexins (natural defense chemicals), and promotes rapid wound healing.
- **Resulting Diseases:** High K application directly reduces the severity of **Stem Rot, Sheath Blight, and Brown Spot** (which is a nutrient-deficiency "poor man's disease").

3. Silica (Si) — The Physical Barrier

- **Effect of Si:** Rice is a silicon-accumulator. Silica deposits in the epidermal cells of leaves, forming a rigid **silicon-cellulose membrane barrier**.
- **Resulting Diseases:** This physical layer prevents direct penetration by fungal appressoria (**Rice Blast, Sheath Blight**) and reduces insect wounding.

4. Phosphorus (P), Magnesium (Mg) & Micronutrients

- Balanced application prevents nutrient deficiency stresses that trigger opportunistic pathogens (e.g. **Brown Spot of Rice** thrives in low N, low K, low Mg soils).

Source: Lecture notes — 3. Rice, Page 3

Q17: How can you control stem rot and ufra of rice? [Paper 6]

Source: Original Course Content

Answer

1. Control of Stem Rot (Sclerotium oryzae)

- **Resistant Cultivars:** Grow resistant varieties like **Biplob, Borishal, Progoti, Mukta**, etc.
- **Field Drainage:** Drain out standing water from infected fields. Allow the soil to dry and crack before re-watering to destroy floating sclerotia.
- **Sanitation:** Burn stubble and plant residues post-harvest to reduce soil-borne sclerotia.
- **Fertilization:** Split N applications and balance with Potash (K).
- **Fungicides:** Spray **Benlate or Homai** @ 0.2% at the booting stage and repeat before flowering.

2. Control of Ufra Disease (Ditylenchus angustus - Nematode)

- **Resistant Cultivars:** Use cultivars with tolerance or resistance, such as **Digha**.
- **Field Sanitation:** Burn the stubble of the previous infected crop completely during the dry summer.
- **Summer Tillage:** Perform deep summer ploughing to expose soil nematodes to direct solar heat.
- **Water Management & Rotation:** Practice crop rotation and avoid planting in low-lying, permanently flooded areas.
- **Chemical Nematicide:** Apply granular nematicide **Furadan (Carbofuran) @ 5 kg/bigha** to infected fields.
- **Seed Treatment:** Soak seeds in a solution of **Carbofuran** at the rate of 0.3 kg/80 kg seeds.

Source: Lecture notes — 3. Rice, Page 4

Q18: Draw and describe the symptoms of blast and bakanae of rice. [Paper 7]

Source: Original Course Content

Answer

1. Rice Blast (Pyricularia oryzae)

- Spindle or eye-shaped leaf lesions with grey/ash-grey centers and dark brown margins.
- Black, rotting nodal joints that lodge (break) easily.
- Rotting neck of the panicle (neck blast) leading to erect empty "white heads" (early) or broken panicles (late).

2. Bakanae / Foot Rot of Rice (Fusarium moniliforme / Gibberella fujikuroi)

- **"Bakanae" means "foolish seedling" in Japanese.**
- **Abnormal Height:** Infected plants are **several inches taller** than normal, healthy plants in the seedbed or field.
- **Appearance:** Plants are thin, lean, pale green to yellowish-green in color, and support few tillers.
- **Adventitious Roots:** The infected plants characteristically produce **adventitious roots at the lower nodes** above the soil level.
- **Lodging/Discoloration:** The lower stem shows straw-colored discoloration. Infected tillers usually die before reaching maturity, and surviving plants produce light, empty, or poor panicles.

(Diagram guide: Draw a normal rice seedling, and next to it draw a bakanae seedling that is much taller, thin, yellow-green, with small adventitious roots emerging from the lower above-ground nodes).

Source: Lecture notes — 3. Rice, Page 2; 7. Scientific Names, Page 1

Q19: Draw and describe the disease cycle of stem rot of rice. [Paper 7]

Source: Original Course Content

Answer

Disease Cycle of Stem Rot of Rice (Sclerotium oryzae)

1. Survival Stage (Overwintering)

- The pathogen is soil-borne and survives in the field as **black, spherical, minute sclerotia** inside infected rice stubbles, crop residues, or loose in the soil.
- These sclerotia can survive in dry soil for up to 6 years and can float on water.

2. Primary Infection

- **Dissemination:** During field preparation and irrigation, the sclerotia float on stagnant water.
- **Penetration:** Floating sclerotia come into contact with the leaf sheaths of rice plants at the water level. Under favorable warm conditions, they germinate and penetrate the leaf sheath directly or through wounds (often caused by stem borers).

3. Disease Development (Mycelial Stage)

- The mycelium invades the leaf sheath, producing black lesions.
- It penetrates the stem (culm) wall, rotting the vascular bundles and parenchyma tissues. The inside of the culm becomes hollow and turns black.

4. Reproduction & Completion of Cycle

- As the crop matures and the tissues dry, the fungus produces **numerous tiny black sclerotia** inside the hollow culm and leaf sheaths.
- The weakened stems collapse (lodge), and during harvest, stubbles containing sclerotia remain in the soil, completing the cycle.

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* graph TD * A[Sclerotia in Stubbles / Soil] -->|Irrigation Water| B[Floating Sclerotia at Water Level] * B -->|Contact & Germination| C[Penetration of Leaf Sheath] * C -->|Mycelial Invasion| D[Lesions expand, Culm rot] * D -->|Fungal Multiplication| E[Hollow black culm / Lodging] * E -->|Harvest / Crop residue| A
```

Source: Lecture notes — 3. Rice, Pages 1, 3, 4

Q20: Write down the control measures of brown spot and stem rot of rice. [Paper 7]

Source: Original Course Content

Answer

Refer to the prescription for Brown Spot and Stem Rot:

1. Control of Brown Spot (Bipolaris oryzae)

- **Cultivar Selection:** Cultivate resistant varieties such as **Biplob, Brrisail, Progoti, Mukta**, etc.
- **Clean Seeds:** Use certified, healthy seeds from disease-free areas.
- **Seed Disinfection:** Perform Hot Water Treatment (50°C for 30 minutes) or treat seeds with **Vitavax-200** @ 0.25% seed weight.
- **Balanced Nutrition:** Apply recommended doses of Nitrogen, Potash, Magnesium, and Silica. Keep the soil wet.
- **Chemical Spray:** Foliar spray of **Benlate or Hinosan** @ 0.2% starting from 60 days of seedling.

2. Control of Stem Rot (Sclerotium oryzae)

- **Cultivar Selection:** Cultivate resistant varieties like **Biplob, Borishal, Progoti, Mukta**, etc.
- **Water Management:** Drain standing water from infected fields. Keep fields dry and cracked before re-watering.
- **Residue Management:** Burn infected rice stubble and trash immediately after harvest to destroy soil-borne sclerotia.
- **Balanced Fertilization:** Balanced nitrogen and high Potash (K) fertilizer.
- **Chemical Spray:** Foliar spray of **Benlate, Homai, or Topsin-M** @ 0.2% at booting and pre-flowering stages.

Source: Lecture notes — 3. Rice, Page 4

Q21: Draw and label the symptoms of Blast, Sheath blight, Bacterial leaf blight, and Bakanae of rice with their causal organisms. [Paper 8]

Source: Original Course Content

Answer

Here are the causal organisms and key visual diagnostic features of these four major rice diseases:

Disease Name	Causal Organism	Diagnostic Symptoms	Labeled Diagram Features
Rice Blast	*Pyricularia oryzae*	Spindle/eye-shaped leaf spots (ash center, dark brown border); black neck rot.	**Spindle-shaped spot** with ash-grey center and dark margins tapering at both ends.
Sheath Blight	*Rhizoctonia solani*	Large oval/elliptical grey-green lesions near water level, resembling **cobra snake skin** with brownish sclerotia.	**Cobra skin lesions** on leaf sheaths near water line with tiny round sclerotia.
Bacterial Leaf Blight (BLB)	*Xanthomonas oryzae* pv. *oryzae*	Yellowish/straw-colored leaf blight starting from tip/margins with **irregular, wavy borders**.	Leaf with **wavy-margined necrotic blight** extending downwards from the tip.
Bakanae	*Fusarium moniliforme*	**"Foolish seedling"**: abnormally tall, thin, yellow-green plants with **adventitious roots at lower nodes**.	**Abnormally tall, thin seedling** with roots emerging from the lower nodes above the ground.

Source: Lecture notes — 3. Rice, Pages 1-2; 7. Scientific Names, Page 1; 8. All Disease Differences, Page 1

Q22: Write down the control measures of stem rot and tungro disease of rice. [Paper 8]

Source: Original Course Content

Answer

1. Control Measures for Stem Rot (Sclerotium oryzae)

- **Resistant Cultivars:** Use varieties like **Biplob, Borishal, Progoti, Mukta**, etc.
- **Water Management:** Drain out standing water. Dry the soil completely until it cracks before re-watering.
- **Crop Hygiene:** Burn stubble and plant residues post-harvest. Practice crop rotation.
- **Chemicals:** Foliar spray of **Benlate, Homai, or Topsin-M** @ 0.2% at booting and before flowering.

2. Control Measures for Rice Tungro Disease (RTSV & RTBV)

- **Resistant Cultivars:** Grow resistant varieties such as **Mala, BR-5, BR-8, BR-10, BR-12, Kalojira**, etc.
- **Vector Control:** Tungro is transmitted by the Green Leafhopper (*Nephotettix virescens*). Apply systemic insecticides like **Malathion** @ 0.2% or Mipcin to control the leafhopper population in the field.
- **Field Sanitation:** Uproot and destroy infected plants immediately at the early stage. Keep fields free of weeds and wild grasses that harbor the virus and vector.
- **Simultaneous Planting:** Practice synchronized planting in the community to disrupt vector migration.

Source: Lecture notes — 3. Rice, Page 4

Q23: How can you manage nemic disease(s) of rice? [Paper 4]

Source: Original Course Content

Answer

The major nematode (nemic) diseases of rice include **Ufra** (caused by *Ditylenchus angustus*) and **White Tip** (caused by *Aphelenchoides besseyi*).

Management Strategy for Rice Nematode Diseases:

1. Cultural and Physical Controls

- **Deep Summer Ploughing:** Plough fields deeply during dry summer months. Exposing the soil to hot sun destroys nematodes and their eggs.
- **Stubble Burning:** Burn crop residues and stubbles completely after harvest, as *Ditylenchus angustus* survives in dry stubbles.
- **Crop Rotation:** Rotate rice with non-host crops like pulses, mustard, or fallow periods to break the nematode life cycle.
- **Sowing Date Adjustments:** Adjust sowing dates to avoid favorable moist/warm conditions that accelerate nematode multiplication.

2. Host Plant Resistance

- Cultivate locally resistant or tolerant varieties such as **Digha** (for Ufra).

3. Chemical Control (Nematicides)

- **Soil Application:** Apply granular nematicides like **Furadan (Carbofuran) @ 5 kg/bigha** into the soil of standing water.
- **Seed Treatment:** Soak seeds in a Carbofuran solution @ 0.3 kg per 80 kg of seed weight prior to sowing.

Source: Lecture notes — 3. Rice, Page 4

Q24: Give the suitable factors for disease development of bakanae of rice. [Paper 4]

Source: Original Course Content

Answer

Bakanae / Foot Rot of Rice (caused by Fusarium moniliforme) is favored by the following environmental and agronomic conditions:

1. High Soil Temperature

- Warm soil temperatures around **35°C** are highly favorable for fungal germination, growth, and secretion of gibberellic acid (which causes abnormal plant elongation).

2. Damp Soil Conditions

- Constant dampness or excessive soil moisture encourages rapid growth and systemic colonization of the fungus inside the coleoptile and root tissues.

3. Excessive Nitrogenous Fertilization

- Applying high amounts of nitrogenous fertilizers creates soft, succulent plants that are easily penetrated by the fungus.

4. Heavy Rainfall and Flooding

- Disseminates the soil-borne conidia and crop residues carrying the fungus across the nursery beds and main fields.

Source: Lecture notes — 3. Rice, Page 3

Q25: Write about the suitable factors for disease development of stem rot and BLB of rice. [Paper 5]

Source: Original Course Content

Answer

1. Favorable Factors for Stem Rot (Sclerotium oryzae)

- **Stagnant Water:** Sclerotia float on stagnant field water, germinating at the water level to infect leaf sheaths.
- **Excessive Nitrogen & Low Potash:** High N yields succulent host tissues; low K reduces the plant's structural strength.
- **Insect Wounds:** Wounds created by **stem borers** provide easy entry ports for the fungus.

2. Favorable Factors for Bacterial Leaf Blight (BLB) (Xanthomonas oryzae pv. oryzae)

- **Warm Temperature:** Favorable temperatures between **25-30°C**.
- **Excess Nitrogen:** Promotes lush, succulent leaf canopy susceptible to bacterial entry.
- **Nutrient Imbalance:** Deficiency of Potash (K) and Phosphorus (P); excess Silicon and Magnesium.
- **Heavy Rainfall & Severe Winds:** Strong winds cause leaves to rub and sustain micro-wounds; heavy rain splashes disperse bacterial cells across the crop canopy.

Source: Lecture notes — 3. Rice, Page 3

Wheat & Barley Diseases

Q1: List different spores produced by the stem rust of wheat. [Paper 4, 5]

Source: Original Course Content

Answer

Black Rust / Stem Rust of wheat (caused by the heteroecious, macrocyclic fungus Puccinia graminis f. sp. tritici) produces five distinct types of spores in a strict chronological sequence during its life cycle:

Spore Stage	Spore Name	Ploidy State	Host Plant	Functional Role
Stage 0	**Pycniospores** (Spermatia)	Haploid (*n*)	Barberry (*Berberis vulgaris*)	Involved in sexual reproduction (spermatization).
Stage I	**Aeciospores**	Dikaryotic (*n*+*n*)	Barberry (*Berberis vulgaris*)	Formed on barberry leaves; carried by wind to infect **wheat**.
Stage II	**Urediniospores** (Uredospores)	Dikaryotic (*n*+*n*)	Wheat (*Triticum aestivum*)	Repeating spore stage. Disseminates and multiplies rust on wheat in summer.
Stage III	**Teliospores**	Dikaryotic to Diploid (2*n*)	Wheat (*Triticum aestivum*)	Overwintering survival stage on dry wheat straw. Undergoes karyogamy.
Stage IV	**Basidiospores**	Haploid (*n*)	Soil / Wheat Straw	Formed from germinating teliospores in spring; wind-blown to infect **barberry**.

Source: Lecture notes — 5. Wheat and Pulse, Pages 1, 3; 7. Scientific Names, Page 3

Q2: Write six diseases of wheat with their causal agents. [Paper 4]

Source: Original Course Content

Answer

Here are six major diseases of wheat along with their causal agents:

#	Disease Name	Causal Agent (Scientific Name)	Pathogen Type
1	**Black Rust / Stem Rust**	*Puccinia graminis* var. *tritici*	Fungus
2	**Leaf Rust / Brown Rust**	*Puccinia recondita* (syn. *Puccinia triticina*)	Fungus
3	**Stripe Rust / Yellow Rust**	*Puccinia striiformis*	Fungus
4	**Loose Smut of Wheat**	*Ustilago tritici* (syn. *Ustilago nuda*)	Fungus
5	**Bipolaris Leaf Blotch / Spot**	*Bipolaris sorokiniana* (Perfect: *Cochliobolus sativus*)	Fungus
6	**Black Point / Seed Smut**	*Bipolaris sorokiniana* / *Fusarium* spp.	Fungal complex

Source: Lecture notes — 7. Scientific Names, Page 3

Q3: List four diseases of wheat with their causal agents. [Paper 5, 6]

Source: Original Course Content

Answer

Here are four common diseases of wheat along with their causal organisms:

1. **Loose Smut of Wheat:** *Ustilago tritici* (Fungus)
2. **Black Rust / Stem Rust:** *Puccinia graminis* var. *tritici* (Fungus)
3. **Bipolaris Leaf Blotch / Spot:** *Bipolaris sorokiniana* (Fungus)
4. **Leaf Rust / Brown Rust:** *Puccinia recondita* (Fungus)

Source: Lecture notes — 7. Scientific Names, Page 3

Q4: Distinguish between loose smut of wheat/barley and covered smut of barley. [Paper 4]

Source: Original Course Content

Answer

Loose Smut and Covered Smut are major fungal diseases of small grain cereals, distinguished by symptoms and pathogen behavior:

Feature	Loose Smut (*Ustilago tritici* / *U. nuda*)	Covered Smut (*Ustilago hordei*)
Ear Emergence	Smutted ears emerge **slightly earlier** than healthy ones.	Smutted ears emerge **later or at the same time** as healthy ones.
Sorus/Spore Mass	Grain is replaced by a **loose, dusty, olive-black mass** of spores.	Grain is replaced by a **hard, enclosed black smut sorus**.
Membrane Cover	Covered by a **delicate, silvery membrane** that ruptures rapidly.	Covered by a **persistent, tough, grey membrane** that resists easy rupturing.
Spore Dispersal	Spores are blown away by wind during flowering, leaving a bare rachis.	Spores remain enclosed until harvest and threshing.
Infection Mode	**Internally seed-borne** (mycelium invades embryonic tissues).	**Externally seed-borne** (spores stick to seed coat during threshing).
Causal Agent	*Ustilago tritici* (Wheat) / *Ustilago nuda* (Barley).	*Ustilago hordei* (Barley).

Source: Lecture notes — 4. Barley,Maize, Cotton and Jute, Pages 1, 3; 8. All Disease Differences, Page 1

Q5: Write down the importance of barberry plant for black rust of wheat. List the different types of spores produced by Puccinia graminis tritici in its disease cycle. [Paper 3, 8]

Source: Original Course Content

Answer

1. Importance of Barberry Plant (Berberis vulgaris) for Black Rust of Wheat

- **Alternate Host:** *Puccinia graminis* is heteroecious (needs two hosts to complete its life cycle). The barberry plant is the vital **alternate host**, while wheat is the primary host.
- **Sexual Reproduction:** The sexual stage (karyogamy and meiosis) of the rust fungus occurs exclusively on the leaves of the barberry plant.
- **Genetic Variation:** Plasmogamy between different mating types occurs on barberry, giving rise to new virulent races of rust through genetic recombination.
- **Primary Inoculum Source:** Aeciospores developed on barberry leaves in spring are blown by wind to infect nearby wheat crops. Discarding barberry plants disrupts this life cycle.

2. Spores Produced by Puccinia graminis var. tritici:

1. **Pycniospores** (Spermatia) - Haploid (*n*), formed on barberry.
2. **Aeciospores** - Dikaryotic (*n+n*), formed on barberry; infects wheat.
3. **Urediniospores** - Dikaryotic (*n+n*), formed on wheat; repeating cycle.
4. **Teliospores** - Dikaryotic to Diploid (*2n*), formed on wheat straw; overwintering.
5. **Basidiospores** - Haploid (*n*), formed on soil/straw; infects barberry.

Source: Lecture notes — 5. Wheat and Pulse, Pages 1, 3

Q6: How can you control bipolaris spot/blotch of wheat? [Paper 5]

Source: Original Course Content

Answer

Control measures for Bipolaris Leaf Blotch / Spot of Wheat (Bipolaris sorokiniana):

1. Host Resistance

- Cultivate locally recommended resistant wheat varieties.

2. Crop Sanitation and Cultural Control

- Burn crop residues, straw, and stubbles after harvest to destroy soil-borne inoculum.
- Practice **Crop Rotation** with non-host crops (e.g. mustard, pulses) for 2-3 years.
- Apply balanced fertilizers. Maintain adequate soil moisture to prevent water stress.

3. Seed Management

- Select healthy, clean seeds from certified disease-free fields.
- **Seed Treatment:** Treat seeds with systemic fungicides like **Vitavax-200** @ 0.25% of seed weight to eliminate seed-borne mycelium.

4. Foliar Fungicide Spray

- Spray systemic triazole fungicides such as **Tilt (Propiconazole) @ 0.04%** or Companion upon first appearance of leaf spots, and repeat at 15-day intervals if wet/cloudy weather persists.

Source: Lecture notes — 5. Wheat and Pulse, Page 4

Q7: Draw and describe the symptoms of bipolaris spot/blotch of wheat. [Paper 8]

Source: Original Course Content

Answer

Symptoms of Bipolaris Leaf Blotch / Spot of Wheat (Bipolaris sorokiniana)

- **Attacked Stages:** Attacks the wheat plant at all growth stages (seedling to grain filling).
- **Seedling Stage (Rot):** Causes pre-emergence and post-emergence rot. Yellowish oval to oblong spots appear on coleoptiles and young leaves. Spots enlarge and turn dark brown.
- **Leaf Spot Phase:**
- Small, yellowish-brown spots appear on leaves.
- Spots enlarge, turning into **oval to oblong spots with dark brown centers**, often surrounded by a light yellow halo.
- In mature crops, leaf lesions merge, forming large **straw-colored necrotic patches** that dry out leaves completely.
- **Black Point (Grain Phase):** The pathogen infects maturing seeds, causing **black point symptoms** (dark brown to black discoloration) on the embryo end of the grain, reducing germination viability.

(Diagram guide: Draw a wheat leaf showing multiple oval-oblong dark spots with yellow margins coalescing to form large dry patches. Also draw a wheat grain showing black discoloration at the lower embryo end).

Source: Lecture notes — 5. Wheat and Pulse, Page 1; 7. Scientific Names, Page 3

Q8: Write down the suitable factors for disease development of sheath blight of rice and brown spot of mustard. [Paper 8]

Source: Original Course Content

Answer

Environmental and cultural factors that promote disease development:

1. Sheath Blight of Rice (Rhizoctonia solani)

- **Warm Temperature and High Humidity:** Temperatures around **28-32°C** and relative humidity of **95% or higher** are highly favorable.
- **Excessive Nitrogenous Fertilization:** High N fertilizer produces dense, soft, and succulent foliage that the fungus easily penetrates.
- **High Planting Density (Close Spacing):** Traps humidity and moisture inside the crop canopy, prolonging leaf wetness.

2. Brown Spot / Alternaria Blight of Mustard (Alternaria brassicae)

- **Foggy and Wet Weather:** Cool temperatures (**15-20°C**), dense morning fog, high moisture, and cloudy skies are highly favorable.
- **Long Dew Period:** Prolonged leaf wetness (>10 hours) is critical for conidial germination.
- **Excessive Nitrogen:** Softens leaf tissue, making it susceptible.
- **Sowing Date:** Crops sown late (after November 25) are highly vulnerable to peak fog periods.

Source: Lecture notes — 3. Rice, Page 3; 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 3

Sugarcane Diseases

Q1: Draw and describe the symptoms of red rot and smut of sugarcane. [Paper 3, 4, 5, 7, 8]

■ *Source: Original Course Content*

Answer

1. Symptoms of Red Rot of Sugarcane (Colletotrichum falcatum)

- **Attacked Stage:** Recognized when canes are nearing maturity.
- **Foliar Symptoms:** The third or fourth leaf from the top turns pale, yellow, and drops slightly. The margins dry up, and eventually the entire crown dries.
- **Midrib Spots:** Dark reddish elongated lesions appear on the **leaf midribs**, which develop grey centers.
- **Internal Stalk Symptoms:** Splitting the cane longitudinally reveals the primary diagnostic feature: **bright red internal tissues with distinctive white spots or transverse bands** across the red area.
- **Cane Quality:** Canes shrink, shrivel, lose weight, and become hollow.
- **Odour:** Emits a sour, acidic, fermented alcoholic odor when split.

2. Symptoms of Smut of Sugarcane (Ustilago scitaminea)

- **Whip-like Shoot:** The terminal bud/growing apex of the sugarcane stalk is transformed into a **long, curved, black whip-like structure**, several feet in length.
- **Silvery Membrane:** The whip is initially covered by a thin, silvery-white membrane which soon ruptures, releasing a dusty, black cloud of chlamydospores.
- **Tillering:** Excessive tillering occurs, producing numerous slender, thin, grassy shoots that are flexible and devoid of leaves.

(Diagram guide: For Red Rot, draw a split sugarcane stem showing red-stained tissues with distinct white horizontal/transverse bands. For Smut, draw a grass-like cane stool where the main shoot terminates in a long, black curved whip).

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Pages 1, 4

Q2: What are the suitable factors for the disease development of red rot and smut of sugarcane? [Paper 3]

■ *Source: Original Course Content*

Answer

1. Favorable Factors for Red Rot of Sugarcane (Colletotrichum falcatum)

- **Poor Soil Drainage:** Soil waterlogging and high moisture favor fungal spore survival and primary infection.
- **Unfavorable Temperatures:** Bud germination in cold, damp soil conditions slows growth and increases infection.
- **Insect Injuries:** Wounds caused by **stalk borers, moth borers, and sugarcane weevils** provide easy entry points for fungal spores.
- **Mechanical Damage:** Injuries to setts during cutting, handling, or planting.
- **Monoculture:** Continuous cultivation of susceptible cultivars.

2. Favorable Factors for Smut of Sugarcane (Ustilago scitaminea)

- **Warm and Dry Climate:** Dry windy conditions followed by rains facilitate spore release and wind dispersal.
- **Temperature:** Temperatures between **25-35°C** favor chlamydospore germination.
- **Infested Seed Setts:** Planting sugarcane setts harvested from smut-infected fields.
- **Ratoon Cropping:** Ratooning an infected field keeps the mycelium active in stubbles.

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 3

Q3: Describe symptoms of red rot of sugarcane and its effect on yield. [Paper 2]

■ *Source: Original Course Content*

Answer

1. Symptoms of Red Rot of Sugarcane (Colletotrichum falcatum)

- Pale, yellow, and drooping upper leaves (third and fourth leaves). Canes wither and shrivel.
- Red elongated lesions on leaf midribs.
- **Longitudinal Split:** Splitting the stalk reveals **red internal tissues showing white transverse spots or bands**.
- **Sour Alcoholic Odour:** Fermented smell due to sugars rotting.

2. Effect of Red Rot on Yield (Socio-Economic Impact)

Red rot is called the "cancer of sugarcane" due to its devastating economic impacts:

- **Sucrose Destruction:** The fungus breaks down sucrose into glucose and fructose, drastically **reducing sugar content** and rendering juice unfit for sugar extraction.
- **Tonnage Loss:** Stalks shrivel, turn hollow, dry up, and lose weight, leading to massive tonnage losses (up to 100% loss in susceptible cultivars).
- **Ratoon Failure:** Ratoon crops fail completely as the pathogen kills the underground stubble.
- **Famine-like Impact:** Causes complete crop failure over large areas, forcing sugar mills to shut down.

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Pages 1, 4

Q4: Write short notes on mosaic diseases of sugarcane. [Paper 2]

■ *Source: Original Course Content*

Answer

Short Notes: Sugarcane Mosaic Disease

1. Etiology (Pathogen)

- **Causal Agent:** *Sugarcane Mosaic Virus* (SCMV), a Potyvirus.

2. Symptoms

- **Leaf Patches:** Typical symptoms appear on young leaves as **oval to elongated pale-yellow/chlorotic patches** of irregular size.
- **Mosaic Pattern:** The patches alternate with green areas, forming a mosaic pattern. It is most visible on young, rapidly unfolding leaves.
- **Internodes & Growth:** Infected canes turn thin, internodes are shortened, and plants are stunted. Red-brown necrotic stripes may appear on stems in severe cases.

3. Transmission (Dissemination)

- **Primary:** Primarily transmitted through **infected seed setts** (cuttings).
- **Secondary:** Transmitted mechanically by sap and by insect vectors, mainly **aphids** (e.g. *Aphis maidis*).

4. Control Measures

- Cultivate virus-free setts from certified nurseries.
- Grow resistant sugarcane varieties.
- Rogue out and destroy infected sugarcane clumps.
- Control insect vectors (aphids) using timely insecticide sprays.

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Pages 1, 4; 7. Scientific Names, Page 2

Q5: Name the cause of sugarcane smut. Describe the symptoms and explain hot water treatment for its control. [Paper 2]

■ *Source: Original Course Content*

Answer

1. Causal Agent of Sugarcane Smut

- **Pathogen:** *Ustilago scitaminea* (Fungus).

2. Symptoms

- The growing apex is transformed into a **long, curved, black whip-like structure**, which can grow several feet.
- Dusty black spore mass (chlamydospores) is released when the enclosing silvery membrane ruptures.
- Excessive tillering occurs, producing thin, flexible, grassy shoots.

3. Hot Water Treatment (HWT) for Control

Hot Water Treatment is a physical method used to eradicate internally seed-borne smut mycelium from sugarcane setts.

Procedure of HWT:

1. **Sett Selection:** Harvest healthy setts from certified, smut-free nurseries.
2. **Water Bath Preparation:** Set up a hot water treatment tank equipped with temperature regulators.
3. **Submersion:** Immerse the setts into the water bath.
4. **Treatment Schedule:** Keep setts submerged at a constant **54°C for 4 hours**. (Note: This long duration completely kills internal smut mycelium without destroying the sugarcane buds).
5. **Post-treatment:** Cool the treated setts and plant them in well-prepared, disease-free soil.

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Pages 1, 4

Q6: How can you prevent/control red rot of sugarcane? [Paper 5, 7]

Source: Original Course Content

Answer

Preventative and curative control measures for Red Rot of Sugarcane (Colletotrichum falcatum):

1. Host Resistance

- Cultivate locally recommended resistant sugarcane varieties, such as **Iswardi-2/54, Iswardi-16, and Iswardi-17**.

2. Crop Sanitation and Hygiene

- Rogue out infected clumps, carry them away, and **burn them completely** to destroy soil-borne inoculum.
- Avoid ratooning in fields that had red rot.

3. Healthy Seed Setts

- Collect setts only from healthy, disease-free nursery plants.
- **Coal-tar Dip:** Dip the cut ends of sugarcane setts in **coal-tar** prior to planting to prevent fungal entry through open wounds.

4. Insect Control

- Apply insecticides to control stalk and shoot borers, preventing borer injury wounds that serve as entry points.

5. Physical Seed Treatment

- Subject setts to Hot Water Treatment (HWT) at **55°C for 10 minutes** before planting to eradicate superficial fungal spores.

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 4

Q7: How can you control the red rot and smut of sugarcane? [Paper 7]

Source: Original Course Content

Answer

1. Control of Red Rot (Colletotrichum falcatum)

- Cultivate resistant varieties (**Iswardi-2/54, Iswardi-16, Iswardi-17**).
- Collect setts from certified disease-free fields.
- Dip the cut ends of setts in coal-tar.
- Burn infected crop debris and stubbles.
- Control insect borers with insecticides.
- HWT at **55°C for 10 minutes**.

2. Control of Smut (Ustilago scitaminea)

- Cultivate resistant sugarcane cultivars.
- Rogue out and destroy smut-infected plants carefully (cover whips with paper bags before uprooting to prevent spore dispersal).
- Collect seed setts from healthy crops.
- **Hot Water Treatment:** Soak setts at **54°C for 4 hours** prior to planting.
- Avoid ratooning infected fields.
- Destroy weed hosts and wild cane grasses.

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 4

Q8: How can you control smut of sugarcane and mosaic of mungbean? [Paper 6]

Source: Original Course Content

Answer

1. Control of Sugarcane Smut (Ustilago scitaminea)

- **Resistant Varieties:** Grow smut-resistant sugarcane cultivars.
- **Hot Water Treatment:** Subject seed setts to **HWT at 54°C for 4 hours** to kill internal mycelium.
- **Sanitation:** Uproot and burn infected cane stools immediately. Avoid ratoon cropping in infected fields.
- **Healthy Seed:** Collect setts from certified nurseries.

2. Control of Mosaic of Mungbean (Mungbean Yellow Mosaic Virus - MYMV)

- **Resistant Varieties:** Cultivate high-yielding resistant cultivars such as **BARI Mung-5, BARI Mung-6, BARI Mash-1, 2, 3**.
- **Vector Control:** The virus is transmitted by Whitefly (*Bemisia tabaci*). Spray systemic insecticides like **Malathion @ 0.2%** or Dimethoate to control the whitefly vector.
- **Uprooting (Roguing):** Uproot and burn virus-infected mungbean plants at the early seedling stage to prevent disease spread.
- **Weed Control:** Destroy wild weed hosts (e.g. *Croton sparsiflorus*) around the field borders.

Source: Lecture notes — 5. Wheat and Pulse, Page 4; 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 4

Q9: How can you control the mosaic and smut of sugarcane? [Paper 8]

■ *Source: Original Course Content*

Answer

1. Control of Sugarcane Mosaic (SCMV)

- **Healthy Setts:** Collect seed setts only from certified, virus-free nurseries.
- **Vector Management:** Spray insecticides timely to control the aphid vectors (*Aphis maidis*) that transmit SCMV.
- **Sanitation:** Rogue out and destroy infected sugarcane clumps. Clean field borders of wild weed grasses.
- **Resistance:** Cultivate mosaic-resistant sugarcane varieties.

2. Control of Sugarcane Smut (Ustilago scitaminea)

- **Seed Treatment:** Subject sugarcane setts to **HWT at 54°C for 4 hours** prior to planting.
- **Sanitation:** Carefully bag and remove smut-whip stools and burn them. Avoid ratoon cropping in infested fields.
- **Host Resistance:** Plant smut-resistant varieties.

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 4

Q10: List two diseases of sugarcane with their causal organisms. [Paper 7]

■ *Source: Original Course Content*

Answer

Two major sugarcane diseases and their causal agents:

1. **Red Rot of Sugarcane:** *Colletotrichum falcatum* (Fungus)
2. **Sugarcane Smut:** *Ustilago scitaminea* (Fungus)

Source: Lecture notes — 7. Scientific Names, Page 2

Jute & Cotton Diseases

Q1: List four diseases of jute and two diseases of cotton with their causal agents. [Paper 3, 6, 7]

Source: Original Course Content

Answer

Here is a list of major fiber crop diseases along with their causal pathogens:

1. Jute Diseases (Four Diseases)

1. **Stem Rot of Jute:** *Macrophomina phaseolina* (Fungus)
2. **Black Band of Jute:** *Botryodiplodia theobromae* (syn. *Diplodia corchori*) (Fungus)
3. **Anthracnose of Jute:** *Colletotrichum corchorum* (Fungus)
4. **Jute Mosaic Disease:** *Jute Mosaic Virus* (JMV)

2. Cotton Diseases (Two Diseases)

1. **Angular Leaf Spot of Cotton:** *Xanthomonas malvacearum* (syn. *Xanthomonas axonopodis* pv. *malvacearum*) (Bacteria)
2. **Fusarium Wilt of Cotton:** *Fusarium oxysporum* f. sp. *vasinfectum* (Fungus)

Source: Lecture notes — 7. Scientific Names, Pages 1-2

Q2: Describe the symptoms of stem rot and black band of jute. [Paper 3, 5, 7]

Source: Original Course Content

Answer

1. Symptoms of Stem Rot of Jute (Macrophomina phaseolina)

- **Growth Stages:** Plants are infected at all growth stages (damping-off in seedlings to mature rotting).
- **Seedling Stage:** Causes pre- and post-emergence damping-off.
- **Mature Stem spots:** Buff-colored to dark-brown lesions appear along the apex, margins, midrib, and petioles of leaves, moving to the stem.
- **Lesions:** Spots on the stem show **irregular margins**. The bark rots and is easily stripped off.
- **Exposed Fibres:** Stem rotting exposes the underlying inner wood and fibers, which decay (canker).
- **Frictional Rub Test:** Rubbing the infected stem between hands **does not** stain fingers black with spores.
- **Spore Type:** Fructification yields single-celled pycnidiospores.

2. Symptoms of Black Band of Jute (Diplodia corchori / Botryodiplodia theobromae)

- **Dense Black Band:** Characterized by a **dense black band** around the stem, usually 2-3 feet above the ground level.
- **Lesions:** Spots have characteristically **regular margins**.
- **Leaves & Stem:** Plants lose all leaves, dry up, and remain standing in the field as black stems.
- **Spherical Pycnidia:** The black band is covered with minute spherical pycnidia.
- **Frictional Rub Test:** On rubbing fingers up and down on the infected stem, the **fingers become stained black** with smutty pycnidiospore powder.
- **Spore Type:** Fructification yields double-celled pycnidiospores.

Source: Lecture notes — 4. Barley,Maize, Cotton and Jute, Pages 2, 4; 8. All Disease Differences, Page 1

Q3: How can you control anthracnose of jute? [Paper 3, 5]

Source: Original Course Content

Answer

Anthracnose of Jute is caused by the fungus Colletotrichum corchorum. Recommended control measures:

1. Seed Sanitation

- Collect seeds from healthy, anthracnose-free jute crops.
- **Seed Treatment:** Treat seeds with systemic fungicides like **Vitavax-200** or Homai @ 0.25% of seed weight to eliminate seed-borne inoculum.

2. Agronomic & Cultural Practices

- Burn and destroy infected crop residues (stems, leaves, pods) after harvest.
- Practice **Crop Rotation** with non-susceptible crops for 2-3 years.
- Balance nitrogenous fertilization with adequate Potash (K).

3. Foliar Fungicide Spray

- Spray the crop with **Dithane M-45** or Mancozeb @ 0.2% at 15-day intervals starting from the first appearance of yellowish-brown depressed cankers.

4. Resistant Cultivars

- Grow resistant varieties recommended for the species:
- For *C. capsularies*: D154, CVL-1, CVE-3, CC-45.
- For *C. olitorius*: O-4, O-9897.

Source: Lecture notes — 4. Barley,Maize, Cotton and Jute, Page 4

Q4: Describe the symptoms of angular leaf spot of cotton. How does this disease disseminate? What are the favourable conditions? [Paper 3, 8]

Source: Original Course Content

Answer

1. Symptoms of Angular Leaf Spot of Cotton (Xanthomonas malvacearum)

- **Cotyledon spots:** Round to irregular water-soaked spots appear on germinating cotyledons.
- **Angular Leaf spots:** Small, water-soaked, light-green spots appear on the underside of leaves. These spots enlarge, turn brown to black, and are strictly bounded by veinlets, giving them a distinct **angular shape**.
- **Black Arm Stage:** Lesions extend along the petioles and stems, turning them black. The stem rots and breaks easily (stem dieback).
- **Boll Rot:** Water-soaked lesions appear on bolls, causing boll rot, rotting of seeds, and staining of cotton lint.

2. Dissemination (Spread)

- **Seed lint/fuzz:** The primary source of inoculum is internally/externally seed-borne bacteria on lint and fuzz.
- **Wind-blown Rain splash:** The main method of secondary spread within the field.
- **Irrigation Water:** Carries bacteria from infected plant debris to healthy roots.
- **Machinery & Cultivation:** Human contact and machinery during weeding.

3. Favorable Conditions

- High temperature (**above 30°C**).
- High relative humidity, persistent morning fog, and wet weather.
- Furrow irrigation or stagnant water in the field.
- Close planting spacing creating a dense humid canopy.

Source: Lecture notes — 4. Barley,Maize, Cotton and Jute, Pages 1-3

Q5: How can you control the angular leaf spot of cotton? [Paper 4, 5, 7]

Source: Original Course Content

Answer

Control measures for Angular Leaf Spot of Cotton (Xanthomonas malvacearum):

1. Healthy Seeds & Sanitation

- Collect seeds only from certified disease-free cotton plants.
- **Acid Delinting:** Treat cotton seeds with concentrated sulfuric acid to destroy external lint-borne bacteria.

2. Cultural Practices

- Burn and decompose all infected plant residues, leaves, and bolls after harvest.
- Practice **Crop Rotation** with non-host crops for at least 2 years.
- Space rows properly to allow sunlight and airflow, reducing canopy humidity.

3. Host Resistance

- Plant locally recommended resistant cotton varieties.

4. Chemical / Defoliation Measures

- Use chemical defoliants late in the season to drop leaves early, which reduces late boll rot and prevents lint staining by bacterial ooze.
- Spray copper fungicides (e.g. Copper Oxychloride @ 0.3%) combined with Streptomycin if symptoms appear early.

Source: Lecture notes — 4. Barley,Maize, Cotton and Jute, Page 3

Q6: Compare and contrast between stem rot and black band of jute. [Paper 4, 7]

Source: Original Course Content

Answer

Comparative diagnostic features:

Diagnostic Feature	Stem Rot (*Macrophomina phaseolina*)	Black Band (*Botryodiplodia theobromae*)
Lesion Margin	Spots have an **irregular margin**.	Spots have a **regular margin**.
Frictional Rub Test	Rubbing the stem **does not** stain fingers black.	Rubbing the stem **stains fingers black** with dusty spores.
Lesion Shape/Band	Irregular rotting patches along the stem.	Forms a continuous **dense black band** around the stem.
Foliar signs	Leaves wilt, bark rots, plant falls over.	Plants shed leaves, stem turns charcoal dry, stands erect.
Causal Agent	*Macrophomina phaseolina*	*Botryodiplodia theobromae* (syn. *Diplodia corchori*)
Pycnidiospores	**Single-celled** pycnidiospores.	**Double-celled** pycnidiospores.
Infection Mode	Soil-borne and seed-borne.	Soil-borne, wound invader.

Source: Lecture notes — 8. All Disease Differences, Page 1

Q7: Draw and describe the disease cycle of stem rot of jute. [Paper 4, 7]

Source: Original Course Content

Answer

Disease Cycle of Stem Rot of Jute (Macrophomina phaseolina)

1. Survival Stage (Primary Inoculum)

- The pathogen is soil-borne and seed-borne.
- It survives as **tiny black sclerotia** in infected crop stubbles/residues in the soil or as dormant mycelium/sclerotia inside the seed coat.

2. Primary Infection

- **Sowing:** When infected seeds germinate, the mycelium infects the radicle and plumule, causing pre-emergence damping-off.
- **Soil Infection:** Sclerotia in the soil germinate under warm, moist conditions (>30°C) and infect the collar region of seedlings, producing basal lesions.

3. Disease Development & Secondary Spread

- **Multiplication:** The mycelium invades the cortical tissues and vascular bundles of the stem. The bark decays, exposing the inner wood.
- **Pycnidia formation:** The pathogen produces numerous spherical fruiting structures called **pycnidia** on stem lesions.
- **Secondary Inoculum:** Pycnidia release single-celled pycnidiospores.
- **Dissemination:** Rain splashes and wind carry pycnidiospores to infect the leaves and stems of healthy adjacent jute plants.

4. Seed and Soil Contamination

- The fungus infects the seed pods (capsules), contaminating the seeds.
- As the infected plants lodge and rot, millions of sclerotia are released back into the soil, completing the cycle.

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* graph TD * A[Sclerotia in Soil / Infected Seed] -->|Germination| B[Basal Rot / Seedling Damping-off] * B -->|Mycelial Colonization| C[Irregular Rotting on Stem & Leaves] * C -->|Fructification| D[Pycnidia produce Pycnidiospores] * D -->|Rain Splash & Wind| E[Secondary Infection on Healthy Plants] * E --> C * C -->|Pod Infection| F[Contaminated Seeds] * C -->|Decomposition| A
```

Source: Lecture notes — 4. Barley,Maize, Cotton and Jute, Pages 2, 4

Q8: Illustrate the symptoms of stem rot and anthracnose of jute. [Paper 6]

Source: Original Course Content

Answer

Both diseases cause stem lesions but have distinct diagnostic features:

1. Symptoms of Stem Rot of Jute (Macrophomina phaseolina)

- Irregular margins on stem lesions.
- Decay of bark, causing the fibers to be exposed and rot.
- Rubbing the stem does not turn fingers black.

2. Symptoms of Anthracnose of Jute (Colletotrichum corchorum)

- **Stalk Spots:** Characterized by **yellowish-brown, water-soaked, depressed spots (cankers)** on the stem.
- **Cankerous Crack:** The spots darken to brown and finally black. The center of the lesion develops distinct cracks, exposing the fibers (**cankerous symptoms**).
- **Foliar Spots:** Necrotic brown spots develop on leaves, and pods (capsules) develop sunken necrotic lesions.

(Diagram guide: For stem rot, draw irregular rotting patches stripping the bark. For anthracnose, draw a stem showing clean-edged depressed oval cankers that are cracked in the center to expose internal fibers).

Source: Lecture notes — 4. Barley,Maize, Cotton and Jute, Pages 2, 4

Q9: How can you control the stem rot of jute and angular leaf spot of cotton? [Paper 6]

Source: Original Course Content

Answer

1. Control of Stem Rot of Jute (Macrophomina phaseolina)

- **Healthy Seeds:** Collect seeds from healthy, disease-free plants.
- **Seed treatment:** Treat seeds with **Vitavax-200** or Homai @ 0.25% of seed weight.
- **Agronomy:** Rotate crops (especially with *C. olitorius*). Burn infected residues and stubbles. Use balanced fertilizers (avoid excess N, apply adequate K).
- **Fungicides:** Spray the crop with **Dithane M-45** or Mancozeb @ 0.2% at 15-day intervals.
- **Resistant Cultivars:** Use BRRI-released resistant varieties (e.g. CVL-1, O-4).

2. Control of Angular Leaf Spot of Cotton (Xanthomonas malvacearum)

- **Clean Seeds:** Use certified disease-free seeds.
- **Seed Delinting:** Treat cotton seeds with concentrated sulfuric acid to destroy external lint-borne bacteria.
- **Sanitation:** Burn and destroy all infected plant residues, leaves, and bolls post-harvest.
- **Crop Rotation:** Practice crop rotation for 2-3 years.
- **Spray:** Spray copper fungicides (e.g., Copper Oxychloride @ 0.3%) combined with Streptomycin upon first appearance.

Source: Lecture notes — 4. Barley,Maize, Cotton and Jute, Pages 3, 4

Q10: List four diseases of fibre crops with their causal agents. [Paper 7]

Source: Original Course Content

Answer

Four major diseases of fiber crops (Jute and Cotton) along with their causal agents:

1. **Stem Rot of Jute:** *Macrophomina phaseolina* (Fungus)
2. **Black Band of Jute:** *Botryodiplodia theobromae* (syn. *Diplodia corchori*) (Fungus)
3. **Anthracnose of Jute:** *Colletotrichum corchorum* (Fungus)
4. **Angular Leaf Spot of Cotton:** *Xanthomonas malvacearum* (Bacteria)

Source: Lecture notes — 7. Scientific Names, Pages 1-2

Pulse & Oilseed Diseases

Q1: Enlist two major diseases each of oilseeds and pulses with causal agents. [Paper 2, 7, 8]

■ *Source: Original Course Content*

Answer

Major diseases of oilseeds and pulses along with their scientific causal pathogens:

1. Major Diseases of Oilseeds (Mustard, Groundnut, Sunflower)

1. **Grey Blight / Alternaria Leaf Spot of Mustard:**
• **Causal Agent:** *Alternaria brassicae* (Fungus)
2. **Leaf Spot of Groundnut (Tikka):**
• **Causal Agent:** *Cercospora arachidicola* (Early stage) & *Cercosporidium personatum* (Late stage) (Fungi)
3. **Leaf Spots of Sunflower:**
• **Causal Agent:** *Alternaria helianthi* (Fungus)

2. Major Diseases of Pulses (Mungbean, Lentil, Chickpea)

1. **Mungbean Yellow Mosaic:**
• **Causal Agent:** *Mungbean Yellow Mosaic Virus* (MYMV) (Virus)
2. **Botrytis Gray Mould of Gram:**
• **Causal Agent:** *Botrytis cinerea* (Fungus)
3. **Foot and Root Rot of Pulses:**
• **Causal Agent:** *Sclerotium rolfsii* or *Fusarium oxysporum* (Fungi)

Source: Lecture notes — 7. Scientific Names, Page 2

Q2: Describe the symptoms of mungbean mosaic disease. [Paper 2]

■ *Source: Original Course Content*

Answer

Symptoms of Mungbean Yellow Mosaic Disease

Mungbean Yellow Mosaic is caused by the Mungbean Yellow Mosaic Virus (MYMV) and is characterized by distinct foliar symptoms:

- **Attacked Stages:** Attacks mungbean plants at all growth stages (seedling to maturity).
- **Mosaic Patches:** Typical symptoms appear on young leaf blades as **bright yellow patches** intercepted by normal green areas.
- **Complete Yellowing:** In highly susceptible cultivars or severe systemic infections, the entire leaf blade becomes completely yellow.
- **Stunting:** Infected plants are shortened (stunted), and their leaf size is reduced.
- **Flower & Pod Loss:** Diseased plants produce smaller and fewer flowers. There is a severe **lack of pod setting**; if pods do develop, they are small, deformed, and contain shrivelled seeds.

Source: Lecture notes — 5. Wheat and Pulse, Page 2

Q3: Distinguish between symptoms of cercospora leaf spot and rust of groundnut. [Paper 2]

Source: Original Course Content

Answer

Diagnostic differences between Cercospora Leaf Spot (Tikka) and Rust of Groundnut:

Diagnostic Feature	Tikka / Cercospora Leaf Spot	Groundnut Rust
Causal Agent	*Cercospora arachidicola* (Early) / *Cercosporidium personatum* (Late).	*Puccinia arachidis* (Fungus).
Lesion Type	Necroticspots (tissue death).	Pustules (uredinia) containing powdery spores.
Visual Appearance	**Circular dark-brown spots** with/without yellow halos.	Small, raised, **orange-brown powdery pustules** on leaves.
Location of Spores	Conidiophores formed on upper leaf surface (early) or lower leaf surface (late).	Powdery orange urediniospores break through the epidermis, mostly on the **lower leaf surface**.
Leaf Behavior	Foliage turns yellow and drops off (defoliation).	Leaves curl, dry up, and remain attached to the plant.
Stem & Pod	Spots also appear on petioles and stems.	Pustules appear on leaf petioles, stems, and pods.

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 2; 8. All Disease Differences, Page 1

Q4: Suggest control measures for foot and root rot of pulses. [Paper 2]

Source: Original Course Content

Answer

Foot and Root Rot of Pulses (caused by soil-borne pathogens like Sclerotium rolfsii or Fusarium oxysporum) requires an integrated control strategy:

1. Cultural Controls

- **Deep Summer Ploughing:** Plough fields deeply during summer. Solar heat destroys resting structures (sclerotia and chlamydospores) in the soil.
- **Crop Rotation:** Rotate pulses with non-host crops like wheat, barley, or mustard for 2-3 years.
- **Soil Liming:** Apply lime (Calcium carbonate) to acidic soils to raise pH, as *Fusarium* species thrive in acidic soils.
- **Sanitation:** Collect and burn infected plant roots and debris post-harvest.

2. Biological Control

- Apply **Trichoderma harzianum** composted with organic manures into the soil prior to sowing to suppress soil pathogens via mycoparasitism.

3. Chemical Seed Treatment

- Treat seeds with systemic fungicides like **Vitavax-200** or Provax @ 0.25% of seed weight (2.5 g/kg seed) to protect germinating seedlings.

Source: Lecture notes — 5. Wheat and Pulse, Pages 2, 4

Q5: Draw and describe symptoms of tikka of groundnut and leaf spot of sunflower. [Paper 3]

Source: Original Course Content

Answer

1. Tikka (Leaf Spot) of Groundnut

- **Causal Agents:** *Cercospora arachidicola* (Early spot) & *Cercosporidium personatum* (Late spot).
- **Early Spot Symptoms:** Reddish-brown, irregular circular spots surrounded by a **prominent, bright yellow halo** on the upper leaf surface.
- **Late Spot Symptoms:** Smaller, darker brown or carbon-black circular spots with **no yellow halo**, with spores forming in concentric rings on the lower leaf surface.

2. Leaf Spot of Sunflower (Alternaria helianthi)

- **Lesions:** Typical spots are **circular to irregular** in shape, dark brown to black.
- **Target Board Pattern:** Spots develop **concentric rings surrounded by a yellow halo**.
- **Coalescence:** Spots coalesce, causing leaf blighting, complete drying, and shedding of leaves. In severe cases, the entire leaf canopy is destroyed.

(Diagram guide: For Tikka, draw early spots with large yellow rings around brown dots. For Sunflower leaf spot, draw large circular spots showing concentric ring patterns similar to a target board).

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 2

Q6: How can you control the leaf spot of sunflower? [Paper 3]

Source: Original Course Content

Answer

Leaf Spot of Sunflower is caused by the fungus Alternaria helianthi. Recommended control measures:

1. Clean Seed Source

- Collect seeds only from healthy, disease-free sunflower crops.

2. Seed Disinfection

- Soak seeds in hot water or treat them with chemical fungicides like **Vitavax-200** @ 0.25% seed weight to eradicate seed-borne conidia.

3. Cultural & Sanitation Practices

- Uproot and burn infected crop residues and trash immediately after harvest.
- Practice proper crop spacing (avoid close planting) to reduce microclimate humidity.

4. Foliar Chemical Spray

- Spray the crop with **0.2% Dithane M-45 (Mancozeb)** or Rovral upon the first appearance of concentric ring leaf spots, and repeat 2-3 times at 15-day intervals.

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 4

Q7: Diagrammatically describe the symptoms of alternaria blight of mustard and cercospora leaf spot of mungbean. [Paper 4]

Source: Original Course Content

Answer

1. Symptoms of Alternaria Blight / Grey Blight of Mustard (Alternaria brassicae)

- **Foliar Spots:** Small, circular spots appear on leaves. Spots enlarge and develop **concentric rings (target board pattern)**. Lesions turn greyish-brown.
- **Stem & Pod Spots:** Dark brown to black circular lesions appear on stems and pods.
- **Sunken Pods:** Spots on pods become sunken.
- **Grain Damage:** Maturing seeds inside pods shrivel and become discolored.

2. Symptoms of Cercospora Leaf Spot of Mungbean (Cercospora canescens)

- **Foliar Spots:** Initial spots appear as small, water-soaked, **pin-headed spots** on leaves.
- **Spot Coloration:** Spots turn brown to reddish-brown, expanding to 1.5 cm in diameter.
- **Center & Halo:** Depending on the cultivar, spots develop a **greyish-white center surrounded by a brown halo**.
- **Defoliation:** Severe infections cause leaves to dry up and drop off.

(Diagram guide: Draw a mustard leaf showing large spots with concentric rings like a target board. Draw a mungbean leaf with multiple tiny pin-head reddish spots, some expanding with white centers).

Source: Lecture notes — 5. Wheat and Pulse, Page 2; 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 2

Q8: Name six diseases of pulse crops with their causal agents. [Paper 4]

Source: Original Course Content

Answer

Here are six common diseases of pulse crops along with their causal agents:

#	Disease Name	Causal Agent (Scientific Name)	Pathogen Type
1	**Foot and Root Rot**	*Sclerotium rolfsii* or *Fusarium oxysporum*	Fungus
2	**Cercospora Leaf Spot**	*Cercospora canescens*	Fungus
3	**Mungbean Yellow Mosaic**	*Mungbean Yellow Mosaic Virus* (MYMV)	Virus
4	**Botrytis Gray Mould**	*Botrytis cinerea*	Fungus
5	**Rust of Pulse**	*Uromyces ciceris-arietini*	Fungus
6	**Anthracnose of Pulse**	*Colletotrichum lindemuthianum*	Fungus

Source: Lecture notes — 7. Scientific Names, Page 2

Q9: Enumerate the symptoms and control measures of stemphylium blight of lentil. [Paper 4]

Source: Original Course Content

Answer

1. Symptoms of Stemphylium Blight of Lentil

- **Causal Agent:** *Stemphylium botryosum* (syn. *Stemphylium vesicarium*) (Fungus).
- **Foliar Spots:** Initial symptoms appear as small, water-soaked, light-brown spots on the leaves.
- **Rapid Blighting:** Leaves quickly turn pale green to yellow, dry up, and drop off.
- **Progression:** Blighting starts from the lower branches and rapidly progresses upwards.
- **Defoliation:** In severe cases, massive defoliation occurs, leaving only bare stems, giving the field a **burnt or scorched appearance**.
- **Pods:** Pod setting is severely reduced; if pods form, they contain shrivelled, low-viability seeds.

2. Control Measures

- **Resistant Varieties:** Grow recommended resistant lentil varieties such as **BARI Masur-4, BARI Masur-5, BARI Masur-6**.
- **Sanitation:** Burn crop residues and stubbles. Practice crop rotation.
- **Seed Treatment:** Treat seeds with systemic fungicides like **Provax-200** or Vitavax-200 @ 0.25% seed weight.
- **Chemical Spray:** Foliar spray of fungicides like **Rovral-50 WP** or Companion @ 0.2% upon first appearance of leaf spots, repeating at 10-15 day intervals.

Source: Lecture notes — 5. Wheat and Pulse, Pages 2, 4; 7. Scientific Names, Page 2

Q10: Draw and describe the symptoms of early and late tikka diseases of groundnut. [Paper 5]

■ *Source: Original Course Content*

Answer

Tikka (Cercospora Leaf Spot) is composed of two distinct fungal phases:

1. Early Tikka Disease (Cercospora arachidicola)

- **Timeline:** Symptoms appear early in the season, about 3-4 weeks after planting.
- **Spots:** Spots are sub-circular or irregular, reddish-brown to dark brown on the upper leaf surface.
- **Yellow Halo:** Characterized by a **prominent, bright yellow halo** surrounding each spot.
- **Conidia:** Fungal conidiophores form mostly on the **upper leaf surface**.

2. Late Tikka Disease (Cercosporidium personatum)

- **Timeline:** Symptoms appear later in the season, about 6-8 weeks after planting.
- **Spots:** Spots are smaller, more circular, and much darker brown or carbon-black.
- **Yellow Halo:** Yellow halo is **usually absent** or very faint.
- **Conidia:** Fungal conidiophores and black spore cushions form mostly on the **lower leaf surface** in concentric rings.

(Diagram guide: Draw two groundnut leaflets. On one leaflet, draw large brown spots with distinct broad yellow halos (Early). On the other leaflet, draw small, solid black spots clustered closely on the lower side without halos (Late)).

Source: Lecture notes — 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 2

Q11: How can you control the cercospora leaf spot and mosaic of pulses? [Paper 7]

Source: Original Course Content

Answer

1. Control of Cercospora Leaf Spot of Pulses (Cercospora canescens)

- **Sanitation:** Burn crop residues and weed hosts after harvest.
- **Crop Rotation:** Rotate pulses with non-host crops for at least 2 years.
- **Chemical Spray:** Spray foliar fungicides like **Bavistin (Carbendazim) @ 0.1%** at 7-10 day intervals for 2-3 times starting from early spotting.
- **Resistant Cultivars:** Sow locally recommended resistant pulse varieties.

2. Control of Mungbean Yellow Mosaic (MYMV)

- **Resistant Cultivars:** Cultivate resistant varieties such as **BARI Mung-5, BARI Mung-6, BARI Mash-1, 2, 3**.
- **Vector Control:** Spray systemic insecticides like **Malathion @ 0.2%** to kill the Whitefly (*Bemisia tabaci*) vector.
- **Roguing:** Uproot and burn infected plants at the early seedling stage.
- **Weed Management:** Clean fields of weed hosts around borders.

Source: Lecture notes — 5. Wheat and Pulse, Page 4

Q12: How can you control botrytis grey mould of gram and cercospora leaf spot of mungbean? [Paper 8]

Source: Original Course Content

Answer

1. Control of Botrytis Grey Mould of Gram (Botrytis cinerea)

- **Cultural Measures:** Avoid excessive vegetative growth by balancing nitrogen. Avoid excessive irrigation. Spacing rows properly to reduce humidity.
- **Deep Summer Ploughing:** Exposes sclerotia to solar heat.
- **Intercropping:** Intercrop chickpea (gram) with linseed or mustard to disrupt canopy wetness.
- **Biological Control:** Apply **Trichoderma** spp. composts to the soil.
- **Chemicals:**
- Treat seeds with Thiram + Carbendazim (1:1) @ 3g/kg.
- Spray the crop with **Captan** @ 5-6 kg/ha at 15-day intervals.

2. Control of Cercospora Leaf Spot of Mungbean (Cercospora canescens)

- **Foliar Spray:** Spray systemic fungicide **Bavistin @ 0.1%** at 7-10 day intervals for 2-3 times.
- **Sanitation:** Burn crop debris.
- **Host Resistance:** Plant resistant mungbean varieties.

Source: Lecture notes — 5. Wheat and Pulse, Page 4

Q13: List six diseases of oil seed crops with their causal organisms. [Paper 8]

Source: Original Course Content

Answer

Six major diseases of oilseed crops along with their scientific causal pathogens:

1. **Grey Blight / Alternaria Leaf Spot of Mustard:** *Alternaria brassicae* (Fungus)
2. **Leaf Spot of Sunflower:** *Alternaria helianthi* (Fungus)
3. **Early Tikka of Groundnut:** *Cercospora arachidicola* (Fungus)
4. **Late Tikka of Groundnut:** *Cercosporidium personatum* (Fungus)
5. **Rust of Sunflower:** *Puccinia helianthi* (Fungus)
6. **Sclerotinia Rot of Mustard:** *Sclerotinia sclerotiorum* (Fungus)

Source: Lecture notes — 7. Scientific Names, Page 2

Q14: Prescribe to control Alternaria leaf spot of mustard and Cercospora leaf spot of mungbean. [Paper 5]

Source: Original Course Content

Answer

Prescription for managing these two major diseases:

1. Prescription for Alternaria Leaf Spot of Mustard (Alternaria brassicae)

- **Early Sowing:** Sow mustard early (before **November 25**) so that the vegetative stage matures before peak winter fog.
- **Sanitation:** Burn and destroy all infected crop residues post-harvest.
- **Crop Rotation:** Practice crop rotation with non-cruciferous crops for 2 years.
- **Seed Treatment:** Treat seeds with **Vitavax-200** @ 0.25% of seed weight before sowing.
- **Vector Control:** Spray **Malathion @ 0.2%** to control aphids, which cause injuries that facilitate fungal entry.
- **Cultural Practice:** Plant in areas that receive early morning sun to accelerate dew drying.

2. Prescription for Cercospora Leaf Spot of Mungbean (Cercospora canescens)

- **Chemical Spray:** Foliar spray **Bavistin (Carbendazim) @ 0.1%** at 7-10 day intervals for 2-3 times starting from early spotting.
- **Hygiene:** Collect and burn infected plant debris. Destroy weed hosts around borders.
- **Resistance:** Plant resistant mungbean varieties (e.g. BARI Mung-5, 6).

Source: Lecture notes — 5. Wheat and Pulse, Page 4; 6. Sugarcane, Mustard, Sunflower, Groundnut, Page 4